QUESTION ANALYSIS

EEE455

VLSI Design and Testing

Question Analysis

Sessions 2020–21 to 2018–19 *3 papers, 85 questions, 128 formulas*

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1. At a glance:

This analysis covers **3 papers** (2020-21, 2019-20, 2018-19), with 85 questions and 489 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Delay and logical effort	79	16 %	13	Path delay (40); Gate delay (13); Best number of stages (12)
Combinational logic styles	72	15 %	14	Stick diagrams (20); Static gates (19); Dynamic logic (14)
Fabrication	67	14 %	12	Lithography (24); Etching (18); Oxidation (11)
Power	48	10 %	8	Dynamic power (24); Activity factor (7); Glitches (6)
Sequential circuits and timing	48	10 %	8	Setup and hold (16); Flip-flops (10); Resettable flip-flops (6)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2020-21	2019-20	2018-19	Term tests	All
MOS transistor theory	**4**	**13**	**6**	**10**	**33**
Threshold voltage	·	·	**6**	**5**	11
Leakage and DIBL	·	**5**	·	**5**	10
I-V equations	·	**8**	·	·	8
C-V characteristics	**4**	·	·	·	4
CMOS inverter	**13**	**7**	**18**	·	**38**
DC characteristics	·	**7**	**8**	·	15
Skew	**7**	·	·	·	7
Beta ratio	**6**	·	·	·	6
nMOS load inverters	·	·	**5**	·	5
Advantages of CMOS	·	·	**5**	·	5
Fabrication	**10**	**20**	**17**	**20**	**67**
Lithography	**8**	**8**	·	**8**	24
Etching	·	**6**	**12**	·	18
Oxidation	·	·	**5**	**6**	11
Epitaxy	·	·	·	**6**	6
Cleaning	·	**6**	·	·	6
Diffusion and implantation	**2**	·	·	·	2
Delay and logical effort	**21**	**25**	**23**	**10**	**79**
Path delay	**12**	**13**	**10**	**5**	40
Gate delay	·	**7**	**6**	·	13
Best number of stages	·	·	**7**	**5**	12
Input ordering	·	**5**	·	·	5
Elmore delay	**5**	·	·	·	5

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Topic and subtopic	2020-21	2019-20	2018-19	Term tests	All
Sizing	4	·	·	·	4
Power	12	16	20	·	48
Dynamic power	12	4	8	·	24
Activity factor	·	·	7	·	7
Glitches	·	6	·	·	6
Leakage	·	6	·	·	6
Sources of power	·	·	5	·	5
Interconnect	·	10	·	·	10
Crosstalk	·	7	·	·	7
Sheet resistance	·	3	·	·	3
Scaling	·	11	6	·	17
Scaling rules	·	5	6	·	11
Moore's law	·	6	·	·	6
Combinational logic styles	24	25	18	5	72
Stick diagrams	6	6	8	·	20
Static gates	·	9	5	5	19
Dynamic logic	3	6	5	·	14
Pseudo-nMOS	8	·	·	·	8
Pass transistors	7	·	·	·	7
CVSL	·	4	·	·	4
Sequential circuits and timing	26	10	12	·	48
Setup and hold	10	·	6	·	16
Flip-flops	10	·	·	·	10
Resettable flip-flops	·	·	6	·	6
Hold time	·	6	·	·	6
FSM	6	·	·	·	6
Cycle time	·	4	·	·	4
Datapath	7	·	6	·	13
Bus architectures	7	·	·	·	7
Adders	·	·	6	·	6
Memories and arrays	13	13	14	·	40
SRAM	10	6	·	·	16
PLA	3	·	7	·	10
DRAM	·	·	7	·	7
ROM	·	7	·	·	7
Verilog, VHDL and design flow	14	·	10	·	24
VHDL	·	·	10	·	10
Verilog	8	·	·	·	8
Design flow	6	·	·	·	6

3. Priority list:

Must know: Delay and logical effort (79), Combinational logic styles (72), Fabrication (67), Power (48), Sequential circuits and timing (48), Memories and arrays (40), CMOS inverter (38), MOS transistor theory (33)

Likely: Verilog, VHDL and design flow (24), Scaling (17), Datapath (13)

Less often: Interconnect (10)

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The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. MOS transistor theory:

33 marks, 6 questions, in 3 of 3 papers. Book: Weste Ch. 2; Kang Ch. 3.

Most often derivation (3 of 6). Subtopics: Threshold voltage; Leakage and DIBL; I-V equations; C-V characteristics.

Formulas:

Subtopic	Formula	Equation	Used in
Threshold voltage (11)	Fermi potential	$\phi_F = \frac{kT}{q} \ln \frac{n_i}{N_A}$	18-19 2(a)
	Depletion charge	$Q_{B0} = -\sqrt{2qN_A\varepsilon_{Si}\lvert 2\phi_F\rvert}$	18-19 2(a)
	Threshold, no body effect	$V_{T0} = \phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$	18-19 2(a)
	Threshold with body effect	$V_T = V_{T0} + \gamma\left(\sqrt{\lvert 2\phi_F\rvert + V_{SB}} - \sqrt{\lvert 2\phi_F\rvert}\right)$	18-19 8(a)
Leakage and DIBL (10)	DIBL	$V_t = V_{t0} - \eta V_{ds}$	18-19 8(b)
	Gate leakage	$I_{gate} = WA\left(\frac{V_{DD}}{t_{ox}}\right)^2 e^{-Bt_{ox}/V_{DD}}$	18-19 8(b)
	DIBL	$V_t = V_{t0} - \eta V_{DS}$	19-20 1(d)
	Subthreshold current	$I_{DS} = I_{ds0}\, e^{\frac{V_{GS}-V_t+\eta V_{DS}}{nv_T}}\left(1 - e^{-V_{DS}/v_T}\right)$	19-20 1(d)
	Subthreshold slope	$S = nv_T \ln 10$	19-20 1(d)
I-V equations (8)	Oxide capacitance	$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$	20-21 1(a), 19-20 2(a)
	Gain factor	$\beta = \mu C_{ox}\frac{W}{L}$	19-20 2(a)
	Current, linear	$I_{DS} = \beta\left(V_{GS} - V_t - \frac{V_{DS}}{2}\right)V_{DS}$	19-20 2(a)
	Current, saturation	$I_{DS} = \frac{\beta}{2}(V_{GS} - V_t)^2$	19-20 2(a)
C-V characteristics (4)	**Gate capacitance**	$C_0 = C_{ox}WL$	20-21 1(a)
	Gate capacitance, cutoff	$C_{gb} \approx C_0,\ C_{gs} = C_{gd} = 0$	20-21 1(a)
	Gate capacitance, linear	$C_{gs} = C_{gd} = \frac{1}{2}C_0,\ C_{gb} = 0$	20-21 1(a)

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Subtopic	Formula	Equation	Used in
	Gate capacitance, saturation	$C_{gs} = \frac{2}{3}C_0,\ C_{gd} = C_{gb} = 0$	20-21 1(a)

Symbols:

ϕ_F = Fermi potential of the substrate (V) ↩

k = Boltzmann constant (J/K)

T = temperature (K)

q = electron charge (C) ↩

n_i = intrinsic carrier concentration (m^{-3}) ↩

N_A = substrate acceptor concentration (m^{-3}) ↩

Q_{B0} = depletion charge per area at $V_{SB} = 0$ (C/m^2) ↩

ε_{Si} = permittivity of silicon (F/m) ↩

V_{T0} = threshold voltage at $V_{SB} = 0$ (V) ↩

ϕ_{GC} = work-function difference (V) ↩

Q_{ox} = oxide charge per area (C/m^2) ↩

V_T = threshold voltage (V) ↩

γ = body-effect coefficient, $\sqrt{2qN_A\varepsilon_{Si}}/C_{ox}$ ($V^{1/2}$) ↩

V_{SB} = source-to-body voltage (V) ↩

V_t = threshold voltage (V) ↩

V_{t0} = threshold at $V_{ds} = 0$ (V) ↩

η = DIBL coefficient (about 0.1) ↩

V_{ds} = drain-to-source voltage (V) ↩

I_{gate} = gate leakage current (A) ↩

W = transistor width (m); gate width (m)

A, B = technology constants

V_{DD} = supply voltage (V) ↩

t_{ox} = oxide thickness (m) ↩

I_{ds0} = current at threshold (A) ↩

n = slope factor (1.3–1.5) ↩

v_T = thermal voltage, kT/q (25.9 mV at 300 K) ↩

S = subthreshold slope (mV/decade) ↩

C_{ox} = gate capacitance per area (F/m^2) ↩

ε_{ox} = permittivity of the oxide (F/m); permittivity of SiO_2, $3.9\varepsilon_0$ (F/m) ↩

β = gain factor (A/V^2) ↩

μ = mobility ($m^2/V \cdot s$) ↩

W, L = width, length (m) ↩

I_{DS} = drain current (A) ↩

V_{GS} = gate-to-source voltage (V) ↩

V_{DS} = drain-to-source voltage (V) ↩

C_0 = full gate capacitance (F) ↩

L = gate length (m) ↩

C_{gb} = gate–body capacitance (F) ↩

C_{gs} = gate–source capacitance (F) ↩

C_{gd} = gate–drain capacitance (F) ↩

Questions:

No.	Paper	Marks	Type	Question
Threshold voltage				
1	18-19 2(a)	6	derivation	Find threshold voltage of MOSFET without body effect.
2	18-19 TT2 8(a)	5	derivation	Find the threshold voltage of MOSFET considering the body effect.
Leakage and DIBL				
3	19-20 1(d)	5	theory	Explain the following non-idealities of MOSFET- (i) Drain induced barrier lowering, (ii) subthreshold leakage. What are the effects of temperature on these non-idealities?
4	18-19 TT2 8(b)	5	theory	Briefly explain the following terms: i) Drain induced barrier lowering, ii) Gate leakage.
I-V equations				
5	19-20 2(a)	8	derivation	Find the equations and I-V characteristics curve of MOSFET current using long-channel model.
C-V characteristics				
6	20-21 1(a)	4	theory	Explain the C-V characteristics of intrinsic MOS gate capacitance in different regions.

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4.2. CMOS inverter:

38 marks, 6 questions, in 3 of 3 papers. Book: Weste Ch. 2; Kang Ch. 5, 6.

Most often theory (4 of 6). Subtopics: nMOS load inverters; DC characteristics; Advantages of CMOS; Beta ratio; Skew.

Formulas:

Subtopic	Formula	Equation	Used in
nMOS load inverters (5)	Saturated load, high level	$V_{OH} = V_{DD} - V_{TL}$	18-19 2(c)
	Saturated load, low level	$k_R\,[2(V_{OH} - V_T)V_{OL} - V_{OL}^2] = (V_{OH} - V_{OL})^2$	18-19 2(c)
DC characteristics (15)	CMOS, region B	$V_{out} = V_{in} + V_t + \sqrt{(V_{DD} - 2V_{in})(V_{DD} - 2V_t)}$	18-19 2(d)
	CMOS, region D	$V_{out} = V_{in} - V_t - \sqrt{(2V_{in} - V_{DD})(V_{DD} - 2V_t)}$	18-19 2(d)
	Switching threshold	$V_M = \dfrac{V_{tn} + r(V_{DD} - \lvert V_{tp}\rvert)}{1 + r}$	18-19 2(d)
	Noise-margin points	$V_{IL} = \dfrac{3V_{DD} + 2V_t}{8}$, $V_{IH} = \dfrac{5V_{DD} - 2V_t}{8}$	18-19 2(d)
	Operating point	$I_{Dn} = \lvert I_{Dp}\rvert$	19-20 3(a)
	Switching threshold	$V_M = \dfrac{V_{tn} + r(V_{DD} - \lvert V_{tp}\rvert)}{1 + r}$, $r = \sqrt{\beta_p/\beta_n}$	19-20 3(a)
	Unity-gain inputs	$V_{IL} = \dfrac{3V_{DD} + 2V_t}{8}$, $V_{IH} = \dfrac{5V_{DD} - 2V_t}{8}$	19-20 3(a)
	Noise margins	$NM_L = V_{IL} - V_{OL}$, $NM_H = V_{OH} - V_{IH}$	20-21 2(b), 19-20 3(a)
Beta ratio (6)	**Switching threshold**	$V_M = \dfrac{V_{DD} - \lvert V_{Tp}\rvert + V_{Tn}\sqrt{\beta_n/\beta_p}}{1 + \sqrt{\beta_n/\beta_p}}$	20-21 2(b)
	Symmetric CMOS V_{IL}	$V_{IL} = \dfrac{3V_{DD} + 2V_T}{8}$, $V_{IH} = \dfrac{5V_{DD} - 2V_T}{8}$	20-21 2(b)
Skew (7)	HI-skew inverter	$g_u = \frac{5}{6}$, $g_d = \frac{5}{3}$	20-21 3(d)
	LO-skew inverter	$g_u = \frac{4}{3}$, $g_d = \frac{2}{3}$	20-21 3(d)

Symbols:

V_{OH} = output high voltage (V)

V_{DD} = supply voltage (V) ↩

V_{TL} = load threshold, with body effect (V) ↩

V_{OL} = output low voltage (V)

V_T = driver threshold (V); threshold of either transistor (V) ↩

V_{out}, V_{in} = output and input voltages (V) ↩

V_t = threshold of either transistor (V) ↩

V_M = switching threshold (V); input where $V_{out} = V_{in}$ (V) ↩

V_{tn}, V_{tp} = nMOS and pMOS thresholds (V) ↩

β_p, β_n = gain factors of the pMOS and nMOS (A/V^2) ↩

V_{IL}, V_{IH} = input low and high limits (V)

I_{Dn}, I_{Dp} = nMOS and pMOS drain currents (A) ↩

V_{in} = input voltage (V) ↩

NM_L, NM_H = low and high noise margins (V)

V_{OL}, V_{OH} = output low and high levels, 0 and V_{DD} (V) ↩

NM_L = low noise margin (V) ↩

NM_H = high noise margin (V) ↩

V_{IL} = lower unity-gain input (V) ↩

V_{IH} = upper unity-gain input (V) ↩

β_n/β_p = beta ratio ↩

g_u = logical effort, rising output ↩

g_d = logical effort, falling output ↩

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Questions:

No.	Paper	Marks	Type	Question
nMOS load inverters				
1	18-19 2(c)	5	theory	Explain operation of saturated load NMOS inverter. What are the disadvantages of this configuration?
DC characteristics				
2	19-20 3(a)	7	derivation repeats 18-19 2(d)	Using I-V characteristics curve of MOSFET, find DC characteristics of CMOS inverter.
3	18-19 2(d)	8	derivation	Find DC characteristics of CMOS inverter using V-I characteristics curve.
Advantages of CMOS				
4	18-19 5(b)	5	theory	Write down the advantages of CMOS.
Beta ratio				
5	20-21 2(b)	6	theory	Briefly explain what happens to the transfer characteristics and noise margins when beta ratio is increased in a CMOS inverter pair.
Skew				
6	20-21 3(d)	7	theory	Explain the inverter width sizing for favoring the rising edge, falling edge, or achieving an unskewed inverter.

4.3. Fabrication:

67 marks, 12 questions, in 3 of 3 papers. Book: Weste Ch. 3; Plummer Ch. 4, 5, 6, 7, 8, 9, 10.

Most often theory (10 of 12). Subtopics: Etching; Oxidation; Lithography; Epitaxy; Cleaning; Diffusion and implantation.

Formulas:

Subtopic	Formula	Equation	Used in
Etching (18)	Plasma dissociation	$CF_4 + e^- \rightarrow CF_3 + F + e^-$	19-20 1(a), 18-19 1(a)
	Silicon etch	$Si + 4F \rightarrow SiF_4 \uparrow$	19-20 1(a), 18-19 1(a)
	Adding O_2	$O + CF_3 \rightarrow COF_2 + F$	19-20 1(a), 18-19 1(a)
	Adding H_2	$H + F \rightarrow HF$	19-20 1(a), 18-19 1(a)
	Plasma ionisation	$e^- + A \rightarrow A^+ + 2e^-$	18-19 1(c)
	Oxide etch (dry)	$SiO_2 + 4F \rightarrow SiF_4 \uparrow + O_2$	19-20 1(a)
	Oxide etch (wet)	$SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O$	19-20 1(a)
	Selectivity	$S = \frac{r_{film}}{r_{under}}$	19-20 1(a)
Oxidation (11)	Deal-Grove fluxes	$F_1 = h(C^* - C_O),\ F_2 = D\frac{C_O - C_I}{x_O},\ F_3 = k_S C_I$	18-19 1(b)
	Oxide growth law	$x_O^2 + Ax_O = B(t + \tau)$	18-19 1(b)
	Thin-oxide limit	$x_O \approx \frac{B}{A}(t + \tau)$	18-19 7(b)
	Thick-oxide limit	$x_O \approx \sqrt{B(t + \tau)}$	18-19 7(b)

Subtopic	Formula	Equation	Used in
Lithography (24)	Lithography resolution	$W_{min} = k_1 \frac{\lambda}{NA}$	19-20 1(c), 18-19 7(a)
Epitaxy (6)	Epitaxy reactions	$SiH_4 \rightarrow Si + 2H_2$; $SiCl_4 + 2H_2 \rightarrow Si + 4HCl$	18-19 7(c)
Cleaning (6)	**RCA clean**	SC-1: NH_4OH:H_2O_2:H_2O = 1:1:5; HF 1:50; SC-2: HCl:H_2O_2:H_2O = 1:1:6	19-20 1(b)

Symbols:

e^- = electron ↩

A = gas atom or molecule ↩

A^+ = its positive ion ↩

S = selectivity ↩

r_{film}, r_{under} = etch rates of the film and of the layer under it (nm/min) ↩

F_1, F_2, F_3 = oxidant fluxes: gas to surface, through the oxide, reacting at the interface ($cm^{-2}s^{-1}$) ↩

h = gas-phase mass-transfer coefficient (cm/s) ↩

C^* = oxidant solubility (cm^{-3}) ↩

C_O, C_I = oxidant concentrations at the oxide surface and at the interface (cm^{-3}) ↩

D = diffusivity in SiO_2 (cm^2/s) ↩

x_O = oxide thickness (cm) ↩

k_S = interface rate constant (cm/s) ↩

N_1 = oxidant molecules per unit volume of oxide (cm^{-3})

t = oxidation time (s) ↩

x_i = starting oxide thickness (cm)

W_{min} = smallest printable feature (m) ↩

k_1 = process factor ↩

λ = wavelength (m) ↩

NA = numerical aperture ↩

Questions:

No.	Paper	Marks	Type	Question
Etching				
1	19-20 1(a)	6	theory	Briefly discuss the chemistries involved in in *[sic: should be "involved in"]* etching. Also explain the effect of adding H_2 and O_2 on etching rate.
2	18-19 1(a)	5	theory	What are the effects of O_2 and H_2 adding on etching?
3	18-19 1(c)	7	theory	Explain how does the plasma sustain. Also explain the chemistries involved in plasma etching.
Oxidation				
4	18-19 1(b)	5	derivation	Discuss the silicon oxidation model mentioning its limiting cases.
5	18-19 TT1 7(b)	6	derivation repeats 18-19 1(b)	Explain the silicon oxidation model mentioning its limiting cases.
Lithography				
6	20-21 5(c)	2	theory	Explain the following: (i) positive mask, (ii) negative mask, (iii) positive photoresist and (iv) negative photoresist.
7	20-21 5(d)	6	theory	Describe the process sequence for patterning SiO_2 as illustrated in the figure 5(d).
8	19-20 1(c)	8	theory repeats 18-19 7(a)	Define photolithography. briefly describe the major steps involved in photolithography.
9	18-19 TT1 7(a)	8	theory	Briefly describe steps of photo-lithography.
Epitaxy				
10	18-19 TT1 7(c)	6	theory	Briefly explain vapor phase epitaxy.
Cleaning				
11	19-20 1(b)	6	theory	What is RCA clean? What are the steps? What contaminants can be removed in this process?
Diffusion and implantation				
12	20-21 1(e)	2	theory	Differentiate between the diffusion process and the ion implantation process in semiconductor fabrication.

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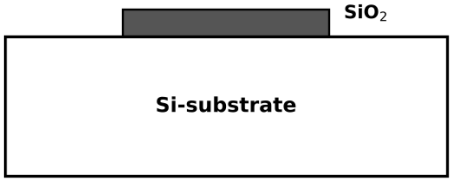


Figure 1: 20-21 5(d): Figure 5(d) of the paper (redrawn)

4.4. Delay and logical effort:

79 marks, 13 questions, in 3 of 3 papers. Book: Weste Ch. 4.

Most often numerical (9 of 13). Subtopics: Gate delay; Path delay; Best number of stages; Input ordering; Elmore delay; Sizing.

Formulas:

Subtopic	Formula	Equation	Used in
Gate delay (13)	**Elmore delay**	$t_{pd} = \sum_i R_{i,\text{to source}}\, C_i$	18-19 3(b)
	NOR3 driving h NOR3	$t_{pdr} = (21+7h)RC$, $t_{pdf} = (9+7h)RC$	18-19 3(b)
	Elmore delay	$t_{pd} = \sum_i R_i C_i$	19-20 2(b)
	NAND3 rising, worst	$t_{pdr} = (9+5h)RC$	19-20 2(b)
	NAND3 falling, worst	$t_{pdf} = (12+5h)RC$	19-20 2(b)
	NAND3 contamination	$t_{cdr} = \left(3+\frac{5h}{3}\right)RC$, $t_{cdf} = (9+5h)RC$	19-20 2(b)
Path delay (40)	Gate delay	$d = gh + p$	20-21 1(d), 19-20 3(b), 18-19 3(c)
	Branching effort	$b = \frac{C_{on} + C_{off}}{C_{on}}$	19-20 3(b), 18-19 3(c)
	Path effort	$F = GBH$	18-19 3(c)
	Minimum delay	$\hat{f} = F^{1/N}$, $D = N\hat{f} + P$	18-19 8(d)
	Sizing	$C_{in} = \frac{g\, C_{out}}{\hat{f}}$	18-19 3(c)
	Logical efforts	inv $g = 1$; NAND-n $\frac{n+2}{3}$; NOR-n $\frac{2n+1}{3}$; AOI22 2	19-20 3(b)
	Path effort	$F = GBH$, $G = \prod g_i$, $B = \prod b_i$, $H = \frac{C_{out}}{C_{in}}$	20-21 1(d), 19-20 3(b)
	Minimum delay	$\hat{f} = F^{1/N}$, $\hat{D} = NF^{1/N} + P$	19-20 3(b)
	Gate size	$C_{in,i} = \frac{g_i C_{out,i}}{\hat{f}}$	20-21 1(d), 19-20 3(b)
	Best stages	$\hat{N} = \log_4 F$	19-20 3(c)
	Logical effort, NAND and NOR	$g_{NAND\text{-}n} = \frac{n+2}{3}$, $g_{NOR\text{-}n} = \frac{2n+1}{3}$	20-21 1(d)

Subtopic	Formula	Equation	Used in
	Minimum path delay	$\hat{D} = NF^{1/N} + P,\ \hat{f} = F^{1/N}$	20-21 1(d)
	Best number of stages	$\hat{N} \approx \log_4 F$	20-21 2(c)
	FO4 delay	$d_{FO4} = 1 \cdot 4 + 1 = 5\tau$	20-21 1(d)
Best number of stages (12)	**Best stage effort**	$p_{inv} + \rho(1 - \ln \rho) = 0$	18-19 8(e)
	Best number of stages	$\hat{N} = \log_\rho F$	18-19 3(d)
Input ordering (5)	Input order (NAND2)	outer $p = 2$, inner $p = 7/3$	19-20 3(d)
Elmore delay (5)	**Elmore delay**	$t_{D,i} = \sum_k R_{ik} C_k$	20-21 2(a)
Sizing (4)	Unit inverter	$W_p = \frac{\mu_n}{\mu_p} W_n$	20-21 3(b)
	Series stack	$W = k\, W_{unit}$	20-21 3(b)
	Path resistance	$\sum_{\text{path}} \frac{R_{unit} W_{unit}}{W_j} \leq R_{unit}$	20-21 3(b)

Symbols:

t_{pd} = propagation delay (s) ↩

$R_{i,\text{to source}}$ = resistance from node i back to the source (Ω) ↩

C_i = capacitance at node i (F) ↩

t_{pdr}, t_{pdf} = rising and falling delays (s)

h = fanout; number of NAND3 loads; electrical effort, C_{out}/C_{in} ↩

R = unit transistor resistance (Ω)

C = unit gate capacitance (F)

R_i = resistance shared with the path to node i (Ω)

t_{pdr} = rising delay (s) ↩

t_{pdf} = falling delay (s) ↩

t_{cdr}, t_{cdf} = rising and falling contamination delays (s) ↩

d = delay in units of τ; delay of one gate (τ) ↩

g = logical effort: inverter 1, NAND2 4/3, NAND3 5/3, NOR2 5/3 ↩

p = parasitic delay: inverter 1, n-input gate n ↩

b = branching effort ↩

C_{on}, C_{off} = capacitance on and off the path (F)

C_{on} = load on the path (F) ↩

C_{off} = load off the path (F) ↩

F = path effort ↩

G = path logical effort, $\prod g_i$ ↩

B = path branching effort, $\prod b_i$ ↩

H = path electrical effort, C_{out}/C_{in} ↩

$\hat{f}$ = best stage effort ↩

N = number of stages ↩

D = path delay ↩

P = path parasitic delay, $\sum p_i$; sum of parasitic delays (τ) ↩

C_{in} = input capacitance of a stage (F)

C_{out} = its load (F)

n = number of inputs ↩

g_i, b_i = logical and branching efforts of stage i

C_{out}, C_{in} = load of the path and input capacitance of its first stage (F) ↩

g_i = logical effort of stage i ↩

b_i = branching of stage i ↩

$\hat{f}$ = best stage effort ↩

$\hat{D}$ = least path delay (τ) ↩

$C_{in,i}$ = input capacitance of stage i (F) ↩

$C_{out,i}$ = load of stage i (F) ↩

$\hat{N}$ = best number of stages ↩

$g_{NAND\text{-}n}$, $g_{NOR\text{-}n}$ = logical efforts of n-input NAND and NOR gates ↩

d_{FO4} = delay of an inverter driving four copies of itself ↩

τ = delay unit of the process (s) ↩

p_{inv} = parasitic delay of an inverter ↩

ρ = best stage effort ↩

$\hat{N}$ = best number of stages ↩

$t_{D,i}$ = delay from the input to node i (s) ↩

C_k = capacitance at node k (F) ↩

R_{ik} = resistance shared by the paths to i and k (Ω) ↩

μ_n, μ_p = electron and hole mobilities ($\text{m}^2/\text{V} \cdot \text{s}$) ↩

W_n = nMOS width ↩

W_p = pMOS width for equal rise and fall ↩

W = width of each transistor in the stack ↩

k = transistors in series on the path ↩

W_{unit} = width in the unit inverter ↩

R_{unit} = resistance of the unit transistor (Ω) ↩

W_j = width of transistor j on the path ↩

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Questions:

No.	Paper	Marks	Type	Question
Gate delay				
1	19-20 2(b)	7	numerical	Find propagation and contamination delay (for both rising and falling) of 3-input NAND gate while it is driving 4 identical NAND gates. Consider that the gate has equal pull-up and pull-down resistance.
2	18-19 3(b)	6	numerical	Sketch a 3-input NOR gate with transistor widths chosen to achieve effective rise and fall resistances equal to a unit inverter. Compute the rising and falling propagation delays of the NOR gate driving h identical NOR gates using the Elmore delay model. Assume that every source or drain has fully contacted diffusion when making your estimate of capacitance.
Path delay				
3	20-21 1(d)	5	numerical	Estimate the minimum delay of the path from A to B in Figure 1(d) and choose transistor sizes to achieve this delay. The initial NAND2 gate may present a load of 8 λ of transistor width on the input and the output load is equivalent to 45 λ of transistor width.
4	20-21 2(c)	7	design	Design an 8-input AND gate using FO4 in different configurations. Determine which configuration provides the fastest performance.
5	19-20 3(b)	6	numerical repeats 18-19 8(d)	Find minimum delay for path $A \rightarrow B$ in the logic circuit of Figure 3.1. Also find the gate size to achieve this delay.
6	19-20 3(c)	7	design	Figure 3.2 shows a portion of a decoder circuit. The inverter at input drives 8 identical 4-input NAND gate with input capacitance y each. The output of the decoder (NAND gate) is connected to a 32-bit word of register and each register bit presents a load of 3-unit sized transistor. Find best number stages (how many inverters should be added between NAND and load?), stage effort, path delay, NAND gate size.
7	18-19 3(c)	10	numerical	For the circuit in figure 3c find- (i) Logical effort, G (ii) Branching effort, B (iii) Electrical effort, E (iv) Path effort, F (v) Stage effort, f (vi) Parasitic delay, P (vii) Minimum path delay, D (viii) Size of the gate.
8	18-19 TT2 8(d)	5	numerical	Find minimum number delay *[sic: should be "minimum delay"]* for the logic circuit below. Also find the gate size to achieve this delay.
Best number of stages				
9	18-19 3(d)	7	derivation	Find the number of stages for minimum delay.
10	18-19 TT2 8(e)	5	numerical	Find the stage effort for the ideal number of stages.
Input ordering				
11	19-20 3(d)	5	theory	Explain input ordering delay effect.
Elmore delay				
12	20-21 2(a)	5	numerical	For the RC circuit shown in Figure 2(a), compute the Elmore delays from input node to node 7 and from input node to node 5 of this circuit in terms of R and C component values.
Sizing				
13	20-21 3(b)	4	numerical	For the logic function $Y = \overline{(A + BC) \cdot D}$, choose transistor widths to achieve equal rise and fall resistance as a unit inverter. Assume, $\mu_n = 3\mu_p$

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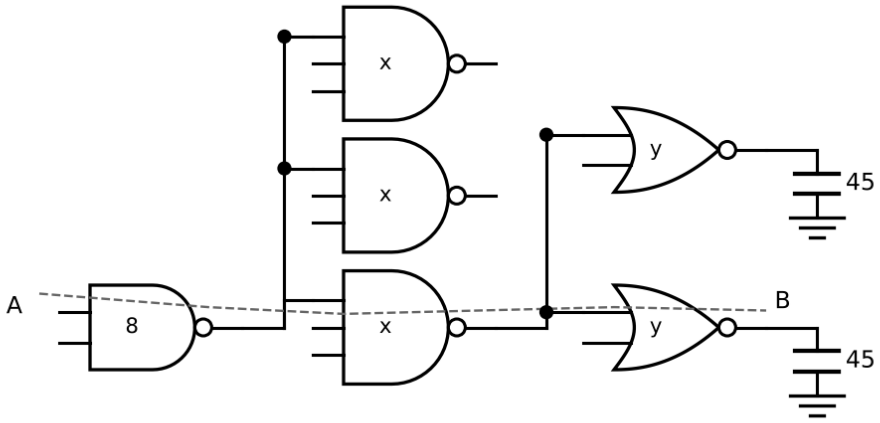


Figure 2: 20-21 1(d): Figure 1(d) of the paper (redrawn)

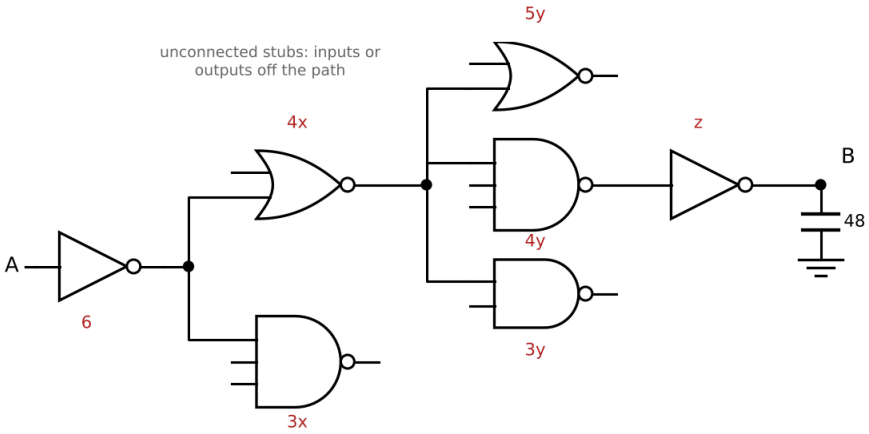


Figure 3: 19-20 3(b): Figure 3.1 of the paper (redrawn)

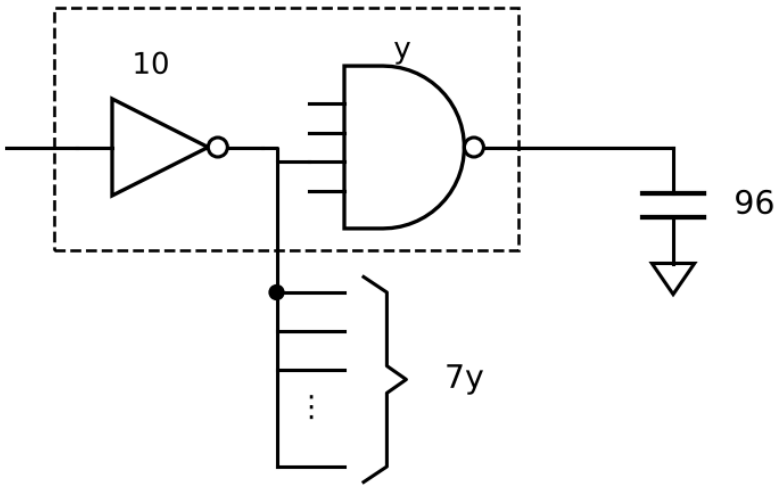


Figure 4: 19-20 3(c): Figure 3.2 of the paper (redrawn)

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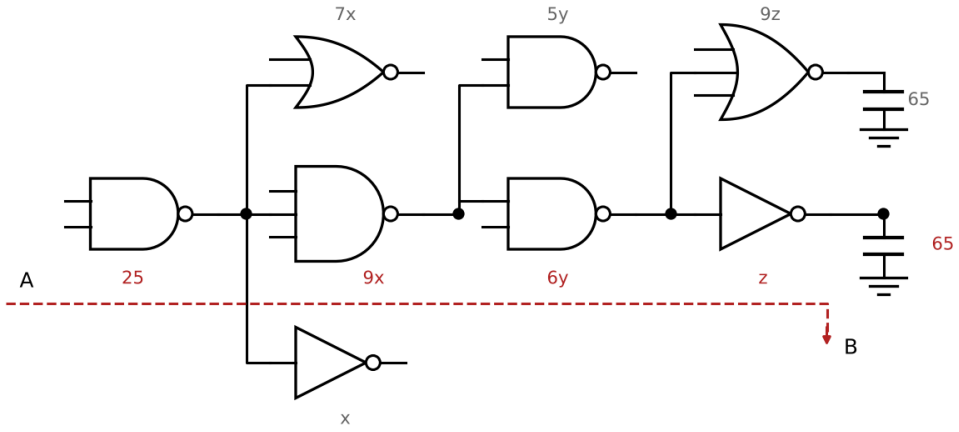


Figure 5: 18-19 3(c): Figure 3c of the paper (redrawn)

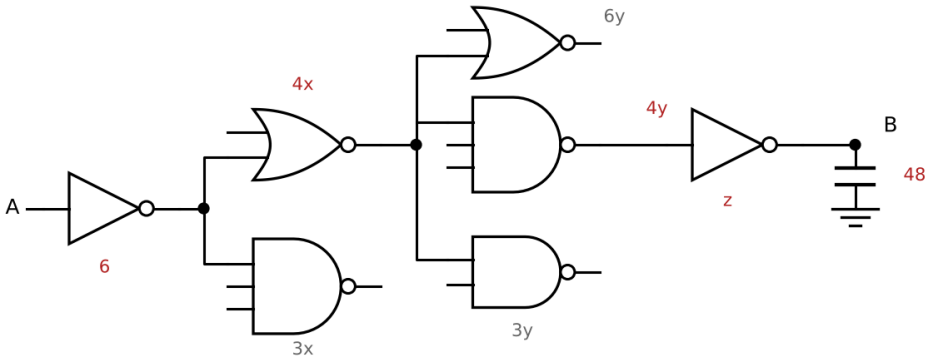


Figure 6: 18-19 TT2 8(d): the circuit of Term Test 2, question 4 (redrawn)

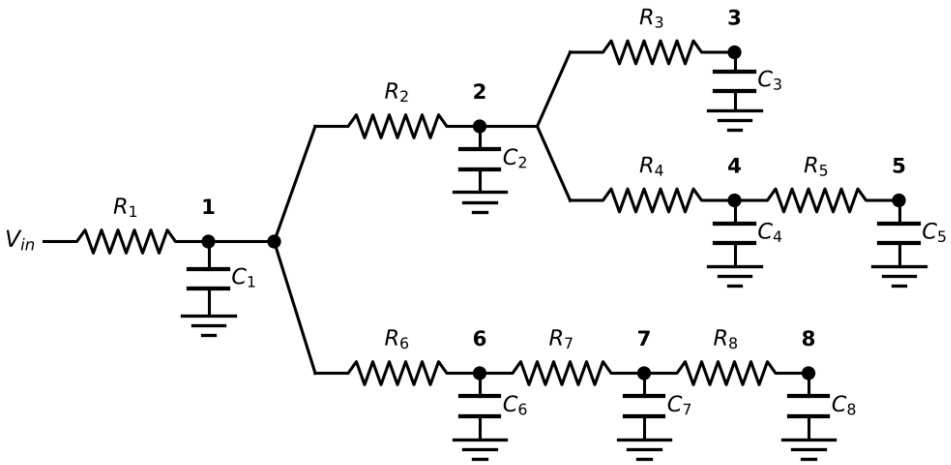


Figure 7: 20-21 2(a): Figure 2(a) of the paper (redrawn)

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4.5. Power:

48 marks, 8 questions, in 3 of 3 papers. Book: Weste Ch. 5.

Most often theory (3 of 8). Subtopics: Sources of power; Activity factor; Dynamic power; Glitches; Leakage.

Formulas:

Subtopic	Formula	Equation	Used in
Sources of power (5)	**Switching power**	$P = \alpha C {V_{DD}}^2 f$	18-19 4(b)
Activity factor (7)	**Activity factor**	$\alpha = P(1-P)$	18-19 4(c)
	Output probabilities	NAND2: $1 - P_A P_B$; NOR2: $\overline{P_A}\,\overline{P_B}$	18-19 4(c)
Dynamic power (24)	Static power	$P_{static} = (I_{sub} + I_{gate} + I_{junct} + I_{contention}) V_{DD}$	18-19 4(d)
	Dynamic power	$P_{dyn} = \alpha C V_{DD}^2 f$	19-20 4(c)
	Capacitance from area	$C = (C/A) \times A$	19-20 4(c)
	Delay from current	$t_{pd} = \dfrac{C_L V_{DD}/2}{I_{sat}} = \dfrac{C_L V_{DD}}{\beta(V_{DD} - V_T)^2}$	20-21 3(c)
	Delay vs supply	$t_{pd} \approx \dfrac{C_L}{\beta V_{DD}} \propto \dfrac{1}{V_{DD}}$	20-21 3(c)
	Energy per output cycle	$E = C_L V_{DD}^2$	20-21 3(c)
	Energy stored	$E_C = \frac{1}{2} C_L V_{DD}^2$	20-21 3(c)
	Dynamic power	$P_{dynamic} = \alpha C_L V_{DD}^2 f$	20-21 2(d)
	Static power	$P_{static} = I_{static} V_{DD}$	20-21 2(d)
Leakage (6)	Stack node voltage	$V_x = \dfrac{\eta V_{DD}}{1 + 2\eta}$	19-20 5(d)
	Stack effect	$\dfrac{I_{single}}{I_{stack}} = 10^{\frac{\eta V_{DD}(1+\eta)}{(1+2\eta)S}}$	19-20 5(d)

Symbols:

P = switching power (W); probability the node is 1 ↩

α = activity factor ↩

C = switched capacitance (F) ↩

V_{DD} = supply voltage (V) ↩

f = clock frequency (Hz) ↩

P_A, P_B = probabilities that inputs A and B are 1 ↩

P_{static} = static power (W) ↩

I_{sub}, I_{gate}, I_{junct}, $I_{contention}$ = subthreshold, gate, junction and contention currents (A) ↩

P_{dyn} = dynamic power (W) ↩

A = chip area (mm^2) ↩

β = gain factor (A/V^2) ↩

V_T = threshold voltage (V) ↩

t_{pd} = propagation delay (s) ↩

C_L = load capacitance (F) ↩

I_{sat} = saturation current (A) ↩

E = energy taken from the supply (J) ↩

E_C = energy in the charged load (J) ↩

$P_{dynamic}$ = switching power (W) ↩

I_{static} = leakage or contention current (A) ↩

V_x = middle node of two OFF nMOS (V) ↩

I_{single}, I_{stack} = leakage of one OFF transistor and of two in series (A) ↩

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Questions:

No.	Paper	Marks	Type	Question
Sources of power				
1	18-19 4(b)	5	theory	What are the sources of power dissipation in CMOS logic circuit?
Activity factor				
2	18-19 4(c)	7	numerical	What is activity factor? Figure 4.1 shows a 4-input AND gate built using a tree and a chain of gates. Determine the activity factors at each node in the circuit assuming the input probabilities PA = PB = PC = PD = 0.5.
Dynamic power				
3	20-21 2(d)	4	theory	Write short note on –i) Static power dissipation ii) Dynamic power dissipation.
4	20-21 3(c)	8	derivation	Show that the dynamic power dissipation of a CMOS inverter varies directly with square of the power supply voltage whereas the delay of the circuit varies inversely with the power supply voltage.
5	19-20 4(c)	4	numerical	You are synthesizing a chip composed of random logic with an average activity factor of 0.1. You are using a standard cell process with an average switching capacitance of 450 pF/mm^2. Estimate the dynamic power consumption of your chip if it has an area of 70 mm^2 and runs at 450 MHz at V_{DD} = 0.9 V.
6	18-19 4(d)	8	numerical	A digital system-on-chip in a 1 V 65 nm process (with 50 nm drawn channel lengths and λ = 25 nm) has 1 billion transistors, of which 50 million are in logic gates and the remainder in memory arrays. The average logic transistor width is 12 λ and the average memory transistor width is 4 λ. The memory arrays are divided into banks and only the necessary bank is activated so the memory activity factor is 0.02. The static CMOS logic gates have an average activity factor of 0.1. Assume each transistor contributes 1 fF/Rm *[sic: should be fF/μm]* of gate capacitance and 0.8 fF/μm of diffusion capacitance. Neglect wire capacitance for now (though it could account for a large fraction of total power). **Estimate the switching power when operating at 1 GHz.** If Subthreshold leakage for OFF devices is 100 nA/μm for low-threshold devices and 10 nA/μm for high-threshold devices. Gate leakage is 5 nA/μm. Junction leakage is negligible. Memories use low-leakage devices everywhere. Logic uses low-leakage devices in all but 5% of the paths that are most critical for performance. **Estimate the static power consumption.**
Glitches				
7	19-20 4(b)	6	theory	Define glitch? How does it increase power consumption? How it can be reduced?
Leakage				
8	19-20 5(d)	6	derivation	Define stack effect. Find subthreshold leakage current for two series OFF transistor.

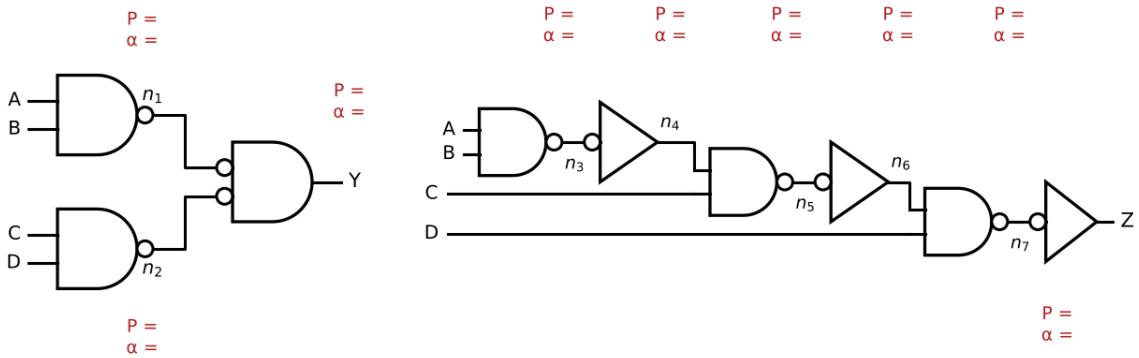


Figure 8: 18-19 4(c): Figure 4.1 (redrawn)

4.6. Interconnect:

10 marks, 2 questions, in 1 of 3 papers. Book: Weste Ch. 6.

Most often theory (2 of 2). Subtopics: Crosstalk; Sheet resistance.

Formulas:

Subtopic	Formula	Equation	Used in
Crosstalk (7)	**Crosstalk, floating victim**	$\Delta V_v = \frac{C_{adj}}{C_{gnd} + C_{adj}} \Delta V_a$	19-20 4(d)
	Crosstalk, driven victim	$\Delta V_v = \frac{C_{adj}}{C_{gnd} + C_{adj}} \cdot \frac{1}{1+k} \Delta V_a,\ k = \frac{\tau_a}{\tau_v}$	19-20 4(d)
	Miller coupling	$C_{eff} = C_{gnd} + k_M C_{adj}$	19-20 4(d)
Sheet resistance (3)	**Sheet resistance**	$R_\square = \frac{\rho}{t},\ R = R_\square \frac{l}{w}$	19-20 4(e)

Symbols:

ΔV_v = noise on the victim (V) ↩

ΔV_a = swing of the aggressor (V) ↩

C_{adj} = coupling capacitance (F) ↩

C_{gnd} = capacitance to ground (F) ↩

τ_a, τ_v = RC time constants of the aggressor and victim drivers (s) ↩

C_{eff} = effective capacitance (F) ↩

k_M = 0 (same), 1 (quiet), 2 (opposite) ↩

$R_\square$ = sheet resistance (Ω/square) ↩

ρ = resistivity (Ω · m) ↩

t = thickness (m) ↩

R = wire resistance (Ω) ↩

l = length (m) ↩

w = width (m) ↩

Questions:

No.	Paper	Marks	Type	Question
Crosstalk				
1	19-20 4(d)	7	theory	Define crosstalk. Show how crosstalk induces noise in different cases. How can crosstalk be reduced?
Sheet resistance				
2	19-20 4(e)	3	theory	Define sheet resistance. How does interconnect affect the performance of a system?

4.7. Scaling:

17 marks, 3 questions, in 2 of 3 papers. Book: Weste Ch. 7.

Most often theory (3 of 3). Subtopics: Scaling rules; Moore's law.

Formulas:

Subtopic	Formula	Equation	Used in
Scaling rules (11)	Moore's law	count ×2 every 2 years	18-19 2(b)
	Constant-field scaling	$L, W, t_{ox}, V : 1/S;\ N_A : S$	18-19 2(b)
	Constant-voltage scaling	$L, W, t_{ox} : 1/S;\ V : 1;\ N_A : S^2$	18-19 2(b)
	Constant-field scaling	$L, W, t_{ox}, V_{DD}, V_t \times \frac{1}{S};\ N_A \times S$	19-20 4(a)
	Constant-voltage scaling	$L, W, t_{ox} \times \frac{1}{S};\ V$ fixed; $N_A \times S^2$	19-20 4(a)

Symbols:

L, W = channel length and width ↩ N_A = substrate doping ↩ V_t = threshold voltage ↩

t_{ox} = oxide thickness ↩ S = scaling factor (> 1) ↩

V = supply voltage ↩ V_{DD} = supply voltage ↩

Questions:

No.	Paper	Marks	Type	Question
Scaling rules				
1	19-20 4(a)	5	theory repeats 18-19 2(b)	What is scaling? Explain how different parameters of transistors are scaled?
2	18-19 2(b)	6	theory	State Moore's law. Explain how different parameters of MOSFET is scaled.
Moore's law				
3	19-20 2(c)	6	theory	State Moore's law. Discuss the problems that arise due to MOSFET scaling.

4.8. Combinational logic styles:

72 marks, 14 questions, in 3 of 3 papers. Book: Weste Ch. 9.

Most often diagram (6 of 14). Subtopics: Static gates; Stick diagrams; Dynamic logic; CVSL; Pseudo-nMOS; Pass transistors.

Formulas:

Subtopic	Formula	Equation	Used in
Static gates (19)	NAND–NAND form	$AB + CD = \overline{\overline{AB} \cdot \overline{CD}}$	19-20 5(c)
	Compound gate	$\text{AOI22} = \overline{AB + CD}$	19-20 5(c)
	Car alarm	$Y = KB\overline{D} + K\overline{B}D = K(B \oplus D)$	20-21 3(a)
Stick diagrams (20)	**OAI22**	$\overline{f} = \overline{(A + B)(C + D)}$	19-20 6(b)
Dynamic logic (14)	Dynamic gate output	$Y = \overline{f}$ in evaluate, $Y = 1$ in precharge	19-20 6(a)
CVSL (4)	CVSL outputs	$Y = f$, $\overline{Y} = \overline{f}$	19-20 2(d)
Pseudo-nMOS (8)	Device gain factor	$k = k'\frac{W}{L}$, $k' = \mu C_{ox}$	20-21 1(b)
	Drain current, linear	$I_D = \frac{k}{2}[2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$	20-21 1(b)
	Drain current, saturation	$I_D = \frac{k}{2}(V_{GS} - V_T)^2$	20-21 1(b)
	Body effect	$V_T = V_{T0} + \gamma\left(\sqrt{\lvert 2\phi_F\rvert + V_{SB}} - \sqrt{\lvert 2\phi_F\rvert}\right)$	20-21 1(b)
	Pseudo-NMOS output low	$k_n[2(V_{DD} - V_{Tn})V_{OL} - V_{OL}^2] = k_p(V_{DD} - \lvert V_{Tp}\rvert)^2$	20-21 1(b)
	Inverter ratio	$\frac{(W/L)_n}{(W/L)_p} = \frac{k_n}{k_p} \cdot \frac{\mu_p C_{ox}}{\mu_n C_{ox}}$	20-21 1(b)
Pass transistors (7)	2-to-1 multiplexer	$Y = \overline{S}D_0 + SD_1$	20-21 6(a)

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Symbols:

A, B, C, D = inputs

Y = alarm output; multiplexer output ↩

K = key in ↩

B = belt fastened ↩

D = door closed ↩

k = device transconductance (A/V^2) ↩

k' = process transconductance (A/V^2) ↩

μ = carrier mobility ($m^2/V \cdot s$) ↩

I_D = drain current (A) ↩

V_{GS} = gate–source voltage (V) ↩

V_{DS} = drain–source voltage (V) ↩

V_T = threshold voltage (V) ↩

V_{T0} = threshold at $V_{SB} = 0$ (V) ↩

γ = body-effect coefficient ($V^{1/2}$) ↩

ϕ_F = Fermi potential (V) ↩

V_{SB} = source–body voltage (V) ↩

V_{Tn}, V_{Tp} = nMOS and pMOS thresholds (V) ↩

V_{OL} = output low voltage (V) ↩

V_{DD} = supply voltage (V) ↩

k_n = driver gain factor (A/V^2) ↩

k_p = load gain factor (A/V^2) ↩

μ_n, μ_p = electron and hole mobilities ($m^2/V \cdot s$) ↩

C_{ox} = gate capacitance per area (F/m^2) ↩

$(W/L)_n$ = driver aspect ratio ↩

$(W/L)_p$ = load aspect ratio ↩

S = select input

D_0, D_1 = data inputs ↩

Questions:

No.	Paper	Marks	Type	Question
Static gates				
1	20-21 3(a)	0	design	You have to design a car alarm system in which the alarm will ring (Y= 1) if anyone of the following condition is satisfied. (i) The key is in (K= 1), The Seat belt is fastened (B=1) and the door is open (D=0) or (ii) The key is in, Door is closed and Seat belt is not fastened.
2	19-20 5(c)	9	diagram	Design a CMOS logic circuit to compute $F = AB + CD$ using (i) only NAND gates, (ii) compound gate. Choose the transistor width to achieve unit path resistance. If each input can present a maximum of 20 λ of transistor width, output must drive a load equivalent to 100 λ of transistor width, calculate the minimum delay to compute F. Which one is faster? Comment on your answer.
3	18-19 4(a)	5	theory	Explain why NMOS network is not used as pull-up network.
4	18-19 TT2 8(c)	5	design	Implement the following expression using CMOS logic: $Y = AB + C(A + B)$.
Stick diagrams				
5	20-21 1(c)	6	diagram	Draw the schematic and stick diagram of the circuit implementing the logic $Y = \overline{(A + B + C) \cdot D}$
6	19-20 6(b)	6	diagram repeats 18-19 5(a)	Draw the stick diagram of the following Boolean function: $f = (A + B).(C + D)$ with their symbolic diagram and truth table.
7	18-19 5(a)	8	diagram	Draw the stick diagram of the following Boolean function: $f = \overline{A.B.C}$ with their symbolic diagram and truth table.
Dynamic logic				
8	20-21 2(e)	3	theory	What is domino gate? Draw the circuit diagram of a domino OR gate.
9	19-20 6(a)	6	theory repeats 18-19 6(a)	Explain the principle of dynamic logic circuit.
10	18-19 6(a)	5	theory	Explain the principle of dynamic inverter.
CVSL				
11	19-20 2(d)	4	theory	Explain the principle of cascode voltage switch logic.

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No.	Paper	Marks	Type	Question
Pseudo-nMOS				
12	20-21 1(b)	8	numerical	A pseudo NMOS inverter is designed such that the output voltage becomes 0.25 V when the input voltage is 5 V. The following data are given: $\mu_n C_{ox}$ = 120 μA/V^2, $\mu_p C_{ox}$ = 50 μA/V^2, V_{ton} = 1V, V_{top} = -1 V, V_{DD} = 5 V, γ = 0.5. Assume that the body of the NMOS transistor is connected with the ground and the body of the PMOS transistor is connected with V_{DD}. (i) Draw the circuit diagram of the inverter. (ii) Calculate the inverter ratio i.e., the ratio of driver transistor (W/L) to the load transistor (W/L). (iii) What will be the output voltage if the input voltage is held at 0.25 V? Explain your answer.
Pass transistors				
13	20-21 4(d)	3	diagram	Implement a 2-input AND gate using both transmission gates and pass-transistor logic.
14	20-21 6(a)	4	diagram	Design a 2-to-1 multiplexer using pass transistor logic. Mention its working procedure.

4.9. Sequential circuits and timing:

48 marks, 8 questions, in 3 of 3 papers. Book: Weste Ch. 10.

Most often theory (3 of 8). Subtopics: Setup and hold; Resettable flip-flops; Hold time; Cycle time; Flip-flops; FSM.

Formulas:

Subtopic	Formula	Equation	Used in
Setup and hold (16)	**Max-delay**	$T_c \geq t_{pcq} + t_{pd} + t_{setup}$	18-19 5(c)
	Min-delay	$t_{cd} \geq t_{hold} - t_{ccq}$	18-19 5(c)
	Latest data arrival	$t_{data,max} = t_{clk1,max} + t_{clk\text{-}q,max} + t_{logic,max}$	20-21 4(a)
	Setup constraint	$t_{data,max} + t_{setup} \leq T + t_{clk2,min}$	20-21 4(a)
	Hold constraint	$t_{data,min} \geq t_{clk2,max} + t_{hold}$	20-21 4(a)
	Maximum frequency	$f_{max} = \frac{1}{T_{min}}$	20-21 4(a)
Resettable flip-flops (6)	Synchronous reset	$Q_{next} = D \cdot \overline{reset}$	18-19 5(d)
Hold time (6)	**Min-delay (hold)**	$t_{cd} \geq t_{hold} - t_{ccq}$	19-20 5(a)
	Hold with skew	$t_{cd} \geq t_{hold} - t_{ccq} + t_{skew}$	19-20 5(a)
Cycle time (4)	**Max-delay (setup)**	$T_c \geq t_{pcq} + t_{pd} + t_{setup}$	19-20 5(b)
	Itanium 2 bypass	$T_c = 90 + 1000 + 62 = 1152$ ps	19-20 5(b)
FSM (6)	Divide-by-3 next state	$S_1' = S_0,\ S_0' = \overline{S_1 + S_0},\ y = \overline{S_1}\,\overline{S_0}$	20-21 6(c)

Symbols:

T_c = clock period (s) ↩

t_{pcq} = clock-to-Q delay of the flip-flop (s); clock-to-Q propagation (s) ↩

t_{pd} = logic propagation delay (s); longest logic delay (s) ↩

t_{setup} = setup time (s) ↩

t_{cd} = logic contamination delay (s); shortest logic delay (s) ↩

t_{hold} = hold time (s) ↩

t_{ccq} = clock-to-Q contamination delay (s) ↩

$t_{data,max}$ = latest arrival at the

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capturing D (s) ↩

$t_{clk1,max}$ = latest launch clock (s) ↩

$t_{clk\text{-}q,max}$ = largest clock-to-Q delay (s) ↩

$t_{logic,max}$ = largest logic and wire delay (s) ↩

T = clock period (s) ↩

$t_{clk2,min}$ = earliest capture clock (s) ↩

$t_{data,min}$ = earliest arrival at the capturing D (s) ↩

$t_{clk2,max}$ = latest capture clock (s) ↩

f_{max} = highest clock frequency (Hz) ↩

T_{min} = least period meeting setup (s) ↩

Q_{next} = state after the clock edge ↩

D = data input ↩

$reset$ = reset input

t_{skew} = clock skew (s) ↩

S_1', S_0' = next-state bits ↩

S_1S_0 = state bits

y = counter output ↩

Questions:

No.	Paper	Marks	Type	Question
Setup and hold				
1	20-21 4(a)	10	numerical	In the following circuit of Figure 4(a), determine whether there is any set up or hold time violation.
2	18-19 5(c)	6	theory	Explain max-delay constraint and min-delay constraint in sequential circuits.
Resettable flip-flops				
3	18-19 5(d)	6	diagram	Sketch logic circuit of flip-flop with synchronous reset and explain its operation.
Hold time				
4	19-20 5(a)	6	theory	What is min-delay constraint? Find min-delay for a system sequenced with flip-flop. If min-delay occurs, what is the solution?
Cycle time				
5	19-20 5(b)	4	numerical	The Itanium 2 ALU self-bypass path is shown in Figure 5.1, and delays associated with each element are given in Table 5.1. Calculate the minimum cycle time T_c at which ALU self-bypass path will operate correctly. The earliest input to the late bypass multiplexer is imm value coming from another flip-flop. Will this path experience any hold-time failures?
Flip-flops				
6	20-21 4(b)	6	diagram	Show the circuit diagram of an asynchronous resettable Flip-Flop and explain its operation utilizing a timing diagram.
7	20-21 6(f)	4	theory	Fill in the truth table for the flip flop shown in Figure 6(f). Is this a positive edge triggered flip flop?
FSM				
8	20-21 6(c)	6	design	You have to design a divide-by-3 counter using FSM. (i) Draw the state transition diagram of the finite state machine described. (ii) Derive the state transition table and equations for the next-state logic. (iii) Draw the complete FSM circuit.

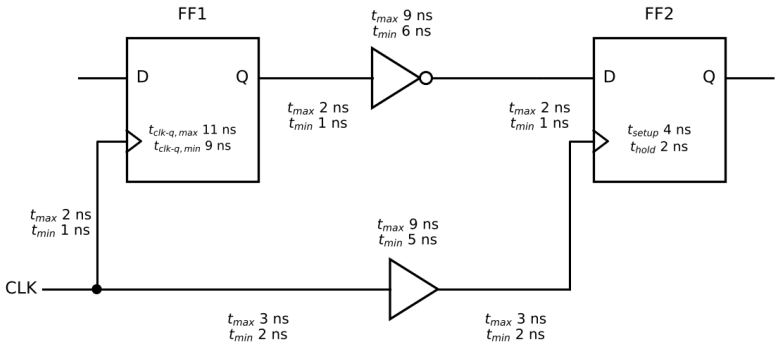


Figure 9: 20-21 4(a): Figure 4(a) of the paper (redrawn)

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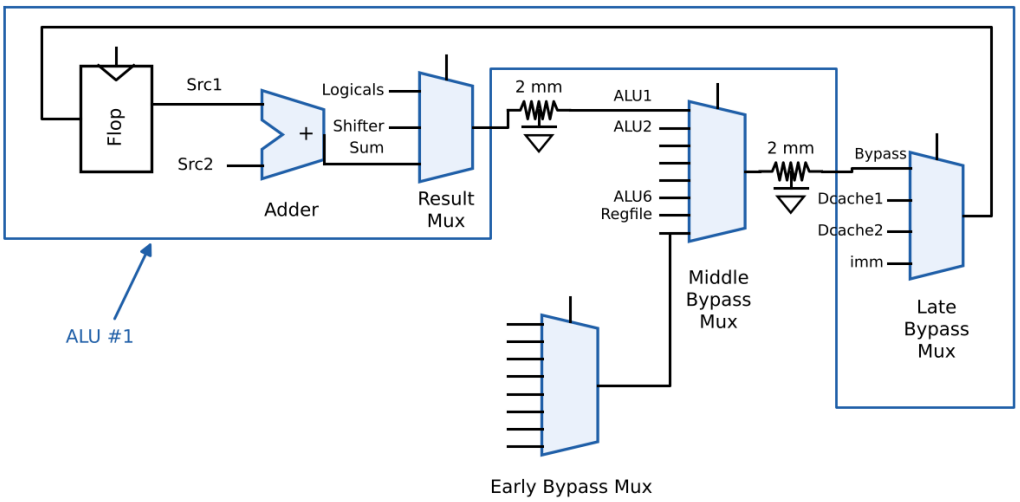


Figure 10: 19-20 5(b): Figure 5.1 of the paper (redrawn)

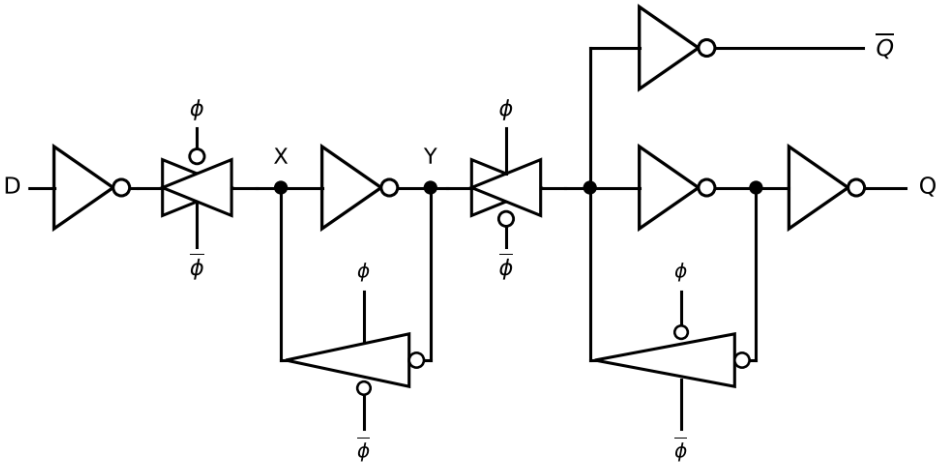


Figure 11: 20-21 6(f): Figure 6(f) of the paper (redrawn)

ϕ	$\overline{\phi}$	D	X	Y	$\overline{Q}$	Q
0	1		1			1
1	0	1				
0	1	1				
1	0	1				

Figure 12: 20-21 6(f): The table of Figure 6(f), as printed

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4.10. Datapath:

13 marks, 2 questions, in 2 of 3 papers. Book: Weste Ch. 11.

Most often design (1 of 2). Subtopics: Adders; Bus architectures.

Formulas:

Subtopic	Formula	Equation	Used in
Adders (6)	**Full adder**	$S = A \oplus B \oplus C_{in}$, $C_{out} = AB + C_{in}(A \oplus B)$	18-19 6(b)
	Mirror adder sum	$S = ABC_{in} + (A + B + C_{in})\overline{C_{out}}$	18-19 6(b)

Symbols:

S = sum ↩ A, B = input bits ↩ C_{in}, C_{out} = carry in and out ↩

Questions:

No.	Paper	Marks	Type	Question
Adders				
1	18-19 6(b)	6	design	Design and implement full adder logic circuit.
Bus architectures				
2	20-21 5(b)	7	theory	A 4-bit datapath consists of register, ALU and a shifter. Show two possible bus architectures of the system such that an addition operation of two operands stored in the register and storing the result back in the register can be computed in at most two clock cycles.

4.11. Memories and arrays:

40 marks, 6 questions, in 3 of 3 papers. Book: Weste Ch. 12.

Most often diagram (4 of 6). Subtopics: PLA; DRAM; SRAM; ROM.

Formulas:

Subtopic	Formula	Equation	Used in
PLA (10)	XOR as sum of products	$A \oplus B = A\overline{B} + \overline{A}B$	20-21 6(e)
DRAM (7)	**DRAM read swing**	$\Delta V = \pm\frac{V_{DD}}{2}\frac{C_{cell}}{C_{cell} + C_{bit}}$	18-19 6(d)
SRAM (16)	SRAM sizing	pull-down > access > pull-up	19-20 6(c)
	Bit-line read time	$t = \frac{C_{bit}\,\Delta V}{I_{cell}}$	20-21 5(a)
ROM (7)	**NOR ROM**	bit line = $\overline{\sum \text{word lines with a transistor}}$	19-20 6(d)

Symbols:

A, B = inputs ↩

ΔV = change of the bit-line voltage (V); swing needed (V) ↩

V_{DD} = supply voltage (V) ↩

C_{cell} = cell capacitance (F) ↩

C_{bit} = bit-line capacitance (F) ↩

t = read time (s) ↩

I_{cell} = cell read current (A) ↩

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Questions:

No.	Paper	Marks	Type	Question
PLA				
1	20-21 6(e)	3	diagram	Sketch a dot diagram for a 2-input XOR using a PLA.
2	18-19 6(c)	7	diagram	Define PLA and PAL. Sketch a 3-input and 2-output PLA implementing the following logic expressions- $s = a\bar{b}\,\bar{c} + \bar{a}b\bar{c} + \bar{a}\,\bar{b}c + abc$, $c = ab + bc + ac$
DRAM				
3	18-19 6(d)	7	theory	Explain how data is written and read in a DRAM sub-array.
SRAM				
4	20-21 5(a)	10	diagram	(i) Show the circuit diagram of a 2 x 2 SRAM array which uses 6-transistor memory cell as basic storage cell. Clearly show the row select, column select, pre-charge and sense signal in your circuit. (ii) Explain how the READ and WRITE operation are performed showing the second row and first column cell as an example. (iii) Explain how sense amplifier reduces the READ time of the cell.
5	19-20 6(c)	6	theory	Draw logic diagram of a SRAM cell and explain its different modes of operation.
ROM				
6	19-20 6(d)	7	diagram	Design and implement a pseudo-nMOS ROM to store the data shown in Figure 6.1. Also draw the dot diagram representation.

4.12. Verilog, VHDL and design flow:

24 marks, 5 questions, in 2 of 3 papers. Book: Weste Appendix A.

Most often design (3 of 5). Subtopics: VHDL; Design flow; Verilog.

Questions:

No.	Paper	Marks	Type	Question
VHDL				
1	18-19 1(d)	8	design	Write the VHDL code to implement the following Boolean expression: $F = AB + \overline{A}BC + BC$.
2	18-19 3(a)	2	theory	What is VHDL? Write down the advantages of VHDL.
Design flow				
3	20-21 4(c)	6	theory	Suppose you have written Verilog code to operate a microprocessor. What will be the upcoming steps till its fabrication? Briefly explain your answer.
Verilog				
4	20-21 6(b)	5	design	Design a 7-segment display driver in Verilog.
5	20-21 6(d)	3	design	Write a Verilog code for the FSM described in Question 6(c).

5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

No.	Paper	Marks	Subtopic	Points to cover
MOS transistor theory				
1	20-21 1(a)	4	C-V characteristics	The gate–oxide–body is a capacitor $C_0 = C_{ox}WL$. On p-type: C_{ox} in accumulation ($V_g < 0$); falls in depletion as the depletion layer adds in series; back to C_{ox} in inversion at low frequency. Across regions the gate capacitance splits into gate–source and gate–drain parts (cutoff, linear, saturation $\frac{2}{3}C_0$ to source).

No.	Paper	Marks	Subtopic	Points to cover
2	19-20 1(d)	5	Leakage and DIBL	DIBL: in a short channel the drain field lowers the source barrier, $V_t = V_{t0} - \eta V_{ds}$, so off current rises with V_{ds}. Subthreshold leakage: below V_t the current falls only exponentially, $I \propto e^{(V_{gs}-V_t)/(nv_T)}$, about 60 to 100 mV per decade. Higher temperature raises both: V_t falls and v_T rises.
3	18-19 TT2 8(b)	5	Leakage and DIBL	DIBL: in a short channel the drain field lowers the source barrier, $V_t = V_{t0} - \eta V_{ds}$, so off current rises with V_{ds}. Gate leakage: electrons tunnel through a thin oxide (below about 15 to 20 Å); high-k dielectrics cut it.
CMOS inverter				
4	20-21 2(b)	6	Beta ratio	Raising β_n/β_p strengthens the nMOS: the VTC moves left, V_M falls ($V_M = \frac{V_{DD}-\lvert V_{tp}\rvert+V_{tn}\sqrt{r}}{1+\sqrt{r}}$), NM_L falls and NM_H rises; they are equal at $\beta_n/\beta_p = 1$.
5	20-21 3(d)	7	Skew	Unskewed: p 2, n 1, $g_u = g_d = 1$. HI-skew (fast rise): p 2, n 1/2, $g_u = 5/6$, $g_d = 5/3$. LO-skew (fast fall): p 1, n 1, $g_u = 4/3$, $g_d = 2/3$.
6	18-19 2(c)	5	nMOS load inverters	Load nMOS with gate tied to drain, always saturated. $V_{OH} = V_{DD} - V_{TL}$ (a threshold drop); V_{OL} set by the driver/load ratio k_R (0.27 V for 5 V, 1 V, $k_R = 9$). Faults: reduced V_{OH}, ratioed V_{OL}, static power when low, slow rise, small gain.
7	18-19 5(b)	5	Advantages of CMOS	Low static power, full swing, ratioless, large noise margins, high input impedance and fan-out, high density, wide supply range.
Fabrication				
8	20-21 1(e)	2	Diffusion and implantation	Diffusion: dopant enters by heat, peak at the surface, more lateral spread, less precise. Ion implantation: accelerated ions, peak below the surface, precise dose and depth, low temperature, needs annealing.
9	20-21 5(c)	2	Lithography	Positive mask: opaque pattern, clear field. Negative mask: clear pattern, opaque field. Positive resist: exposed part dissolves (same image as the mask). Negative resist: exposed part hardens and stays (reverse image).
10	20-21 5(d)	6	Lithography	Clean, thermal oxidation, spin resist and soft bake, align and UV exposure, develop and hard bake, etch the exposed SiO_2 (HF or plasma), strip the resist, leaving the SiO_2 pattern.
11	19-20 1(a)	6	Etching	Etching removes unprotected material. Wet: $SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O$, isotropic and selective. Dry: plasma radicals (CF_4 gives F) form volatile SiF_4; ions make it anisotropic. O_2 in CF_4 removes CF_x, freeing F: the Si rate rises to a peak near 12 % O_2, then falls (dilution, surface oxidation). H_2 ties F up as HF and leaves polymer on Si: the Si rate falls, SiO_2 selectivity rises.
12	19-20 1(b)	6	Cleaning	RCA clean: SC-1 (NH_4OH:H_2O_2:H_2O 1:1:5, 75 to 80 °C) removes organics, particles and some metals; HF dip removes the oxide; SC-2 (HCl:H_2O_2:H_2O 1:1:6) removes alkali ions and metals; each followed by a DI-water rinse.
13	19-20 1(c)	8	Lithography	Photolithography transfers a mask pattern into photoresist with light. Clean, dehydration bake and HMDS prime; spin coat; soft bake; align and expose; post-exposure bake; develop; hard bake; inspect; then etch or implant and strip.
14	18-19 1(a)	5	Etching	O_2 in CF_4 removes CF_x, freeing F: the Si rate rises to a peak near 12 % O_2, then falls (dilution, surface oxidation). H_2 ties F up as HF and leaves polymer on Si: the Si rate falls, SiO_2 selectivity rises.
15	18-19 1(c)	7	Etching	RF-accelerated electrons ionise atoms ($e + A \rightarrow A^+ + 2e$); ions hit the electrodes and free secondary electrons; the plasma lasts while ionisation balances loss. Etch chemistry: radicals (F, Cl) form volatile products (SiF_4, $AlCl_3$, CO_2) that are pumped away; ion bombardment makes it directional.

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No.	Paper	Marks	Subtopic	Points to cover
16	18-19 TT1 7(a)	8	Lithography	Clean, dehydration bake and HMDS prime; spin coat; soft bake; align and expose; post-exposure bake; develop; hard bake; inspect; then etch or implant and strip.
17	18-19 TT1 7(c)	6	Epitaxy	CVD growth of a single-crystal layer that continues the substrate: $SiH_4 \rightarrow Si + 2H_2$ or $SiCl_4 + 2H_2 \rightarrow Si + 4HCl$ at 1000 to 1250 °C on a clean surface; dopants (AsH_3, PH_3, B_2H_6) added to the gas; growth limited by surface reaction or mass transport.
Delay and logical effort				
18	19-20 3(d)	5	Input ordering	The delay depends on which series input switches last: the outer input (next to the output) is faster because the inner node is already discharged ($p = 2$ against 7/3 for a NAND2); put the latest-arriving signal on the outer input.
Power				
19	20-21 2(d)	4	Dynamic power	Static: $P = I_{static}V_{DD}$ from subthreshold, gate and junction leakage and from contention in ratioed logic. Dynamic: $P = \alpha C_L V_{DD}^2 f$ from charging the load, plus short-circuit current while switching.
20	19-20 4(b)	6	Glitches	A glitch is a spurious transition caused by unequal input arrival times. It charges and discharges node capacitance needlessly (CV_{DD}^2 each, a higher α) and propagates. Reduce by balancing paths, shallower logic or pipelining, restructuring, and gating.
21	18-19 4(b)	5	Sources of power	Dynamic: switching power $\alpha CV_{DD}^2 f$ and short-circuit current while both transistors conduct. Static: subthreshold, gate and junction leakage, and contention current in ratioed circuits.
Interconnect				
22	19-20 4(d)	7	Crosstalk	Crosstalk: coupling through C_{adj} from an aggressor to a victim. Floating victim: $\Delta V = \frac{C_{adj}}{C_{gnd}+C_{adj}}\Delta V_{agg}$; driven victim: smaller by $1/(1+k)$ and decays; both switching: effective $C = C_{gnd} + k_M C_{adj}$ (k_M 0 to 2). Reduce by spacing, shielding, buffering, careful routing.
23	19-20 4(e)	3	Sheet resistance	$R_\square = \rho/t$ (Ω per square), $R = R_\square l/w$. Interconnect adds RC delay (growing as length squared), dynamic power, crosstalk and IR-drop noise, electromigration and area.
Scaling				
24	19-20 2(c)	6	Moore's law	Moore's law: transistor count doubles about every two years. Scaling problems: short-channel effects (DIBL, V_t roll-off, punch-through), subthreshold and gate leakage, velocity saturation, interconnect RC delay, power density (end of Dennard scaling), variability, lithography limits.
25	19-20 4(a)	5	Scaling rules	Scaling shrinks all dimensions by $1/S$ per generation. Constant field: L, W, t_{ox}, x_j, V_{DD}, V_t by $1/S$, doping by S; current and capacitance $1/S$, delay $1/S$, power per gate $1/S^2$, density S^2, power density constant. Constant voltage: delay $1/S^2$, power density S^3.
26	18-19 2(b)	6	Scaling rules	Moore's law: transistors per chip double about every two years. Constant-field scaling by $1/S$: lengths, t_{ox}, V_{DD}, V_t by $1/S$, doping by S; current and capacitance $1/S$, delay $1/S$, power per gate $1/S^2$, density S^2, power density constant. Constant-voltage scaling: delay $1/S^2$ but power density rises as S^3.
Combinational logic styles				
27	20-21 1(c)	6	Stick diagrams	nMOS: A, B, C in parallel, in series with D; pMOS: A–B–C in series, in parallel with D. Stick diagram with the common Euler path A, B, C, D.
28	20-21 2(e)	3	Dynamic logic	A domino gate is a dynamic gate (precharge pMOS, nMOS network, clocked foot) followed by a static inverter: non-inverting and safe to cascade. Domino OR: A and B nMOS in parallel under the precharged node X; the inverter gives $Y = A + B$.

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No.	Paper	Marks	Subtopic	Points to cover
29	20-21 4(d)	3	Pass transistors	Transmission gates: pass A when B = 1, pass 0 when B = 0 (full swing). Pass transistors: nMOS(B) passes A, nMOS($\overline{B}$) pulls Y to 0; a weak 1 of $V_{DD} - V_t$ needs restoring.
30	20-21 6(a)	4	Pass transistors	nMOS with gate $\overline{S}$ passes D_0, nMOS with gate S passes D_1, outputs joined: $Y = \overline{S}D_0 + SD_1$; exactly one switch conducts; transmission gates give full swing.
31	19-20 2(d)	4	CVSL	Dual-rail logic: complementary nMOS networks for f and $\overline{f}$ under two cross-coupled pMOS; the conducting network pulls one output low, which turns on the pMOS of the other output; full swing, no static current, both polarities, but more transistors and contention while switching.
32	19-20 6(a)	6	Dynamic logic	One clocked pMOS precharges the output to V_{DD} while $\phi = 0$ (foot off); when $\phi = 1$ the nMOS network evaluates and discharges the output if it conducts, else the output stays high on its capacitance. Fewer transistors ($N + 2$), lower input capacitance, faster; needs monotonic inputs, charge sharing and leakage keepers.
33	19-20 6(b)	6	Stick diagrams	$\overline{f}$ from an OAI22 (nMOS: (A ‖ B) in series with (C ‖ D); pMOS: (A–B) ‖ (C–D)), then an inverter gives f. Euler order A, B, D, C on both strips; $f = 1$ in 9 of 16 rows.
34	18-19 4(a)	5	Static gates	An nMOS pull-up has its source at the output: $V_{GS} = V_{DD} - V_{out}$, so it turns off at $V_{out} = V_{DD} - V_t$: a degraded 1, worse with body effect, slow, and it leaves static current in the next gate. A pMOS passes a strong 1.
35	18-19 5(a)	8	Stick diagrams	$f = \overline{ABC}$ is a NAND3: three parallel pMOS (width 2), three series nMOS (width 3); $f = 0$ only for $A = B = C = 1$. Stick diagram with gate order A, B, C on one Euler path.
36	18-19 6(a)	5	Dynamic logic	Precharge ($\phi = 0$): clocked pMOS charges Y to V_{DD}, foot off. Evaluate ($\phi = 1$): if A = 1, Y discharges; if A = 0, Y stays high on its capacitance; $Y = \overline{A}$. Inputs must be monotonic; leakage needs a keeper.
Sequential circuits and timing				
37	20-21 4(b)	6	Flip-flops	Master–slave transmission-gate flip-flop with a NAND ($\overline{R}$) in the master loop and a NOR (R) in the slave loop; $R = 1$ forces the state to 0 at once, independent of the clock.
38	20-21 6(f)	4	Flip-flops	Rows (X, Y, $\overline{Q}$, Q) = (1, 0, 0, 1), (1, 0, 1, 0), (0, 1, 1, 0), (0, 1, 0, 1); master transparent at $\phi = 0$, slave at $\phi = 1$: yes, positive edge triggered.
39	19-20 5(a)	6	Hold time	Min-delay (hold): the shortest path must not let new data reach the next element within its hold time, $t_{cd} \geq t_{hold} - t_{ccq}$ (plus skew); independent of the clock period. Fix: pad short paths with buffers, use flip-flops with small hold time.
40	18-19 5(c)	6	Setup and hold	Max-delay (setup): $T_c \geq t_{pcq} + t_{pd} + t_{setup}$; a failure is fixed by a slower clock. Min-delay (hold): $t_{cd} \geq t_{hold} - t_{ccq}$; independent of the clock, a failure needs a redesign (add delay).
41	18-19 5(d)	6	Resettable flip-flops	An AND gate on D with $\overline{reset}$ before a master-slave flip-flop: $Q_{next} = D \cdot \overline{reset}$ at the rising edge, so reset acts only at a clock edge.
Datapath				
42	20-21 5(b)	7	Bus architectures	Two buses: cycle 1 reads both operands and adds (result latched), cycle 2 writes back: 2 cycles. Three buses: read A and B and write the result on bus C in 1 cycle.
Memories and arrays				
43	20-21 5(a)	10	SRAM	2 × 2 array of 6T cells; word lines from the row decoder; column-select pass transistors; precharge pMOS on every bitline; sense amplifier and write driver. Read: precharge, WL high, the 0 side discharges a bitline, sense. Write: drive the bitlines, WL high, the cell flips.

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No.	Paper	Marks	Subtopic	Points to cover
44	20-21 6(e)	3	PLA	AND plane: $P_1 = A\overline{B}$, $P_2 = \overline{A}B$; OR plane: $Y = P_1 + P_2$.
45	19-20 6(c)	6	SRAM	6T SRAM: cross-coupled inverters with two access nMOS on WL to bit and $\overline{bit}$. Hold: WL = 0, the latch keeps the data. Read: bitlines precharged, WL = 1, the 0 side discharges its bitline, the sense amplifier reads (pull-down stronger than access). Write: bitlines driven, access stronger than pull-up.
46	19-20 6(d)	7	ROM	2:4 decoder drives 4 word lines; 6 bit lines with grounded-gate pMOS pull-ups and output inverters; an nMOS at each stored 1 (12 transistors); the dot diagram marks the same places.
47	18-19 6(c)	7	PLA	PLA: programmable AND and OR planes. PAL: programmable AND plane, fixed OR. For s and c: 3 inputs (6 lines), 7 product lines, 2 outputs: four products to s, three to c.
48	18-19 6(d)	7	DRAM	Write: the driver forces the bitline, the wordline rises, the cell capacitor takes the level. Read: bitlines precharged to $V_{DD}/2$, wordline high, charge sharing gives $\Delta V = \pm(V_{DD}/2)C_{cell}/(C_{cell} + C_{bit})$, the sense amplifier compares with the reference bitline and restores the full level (destructive read).
Verilog, VHDL and design flow				
49	20-21 4(c)	6	Design flow	RTL verification, logic synthesis (netlist, scan), floorplan, placement and clock tree, routing, sign-off (STA, DRC, LVS, power), GDSII tape-out, masks, wafer fabrication, test, dicing, packaging.
50	18-19 3(a)	2	VHDL	VHDL: IEEE-standard language that describes digital hardware for simulation and synthesis. Advantages: portable, several levels of description (behavioural to structural), verification before fabrication, faster design through synthesis, reusable.

6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
MOS transistor theory				
1	19-20 2(a)	8	I-V equations	Channel charge $C_{ox}W(V_{GS} - V_t - V)$; speed $\mu_n\, dV/dx$; $I = Qv$; integrate source to drain for the linear law; put $V_{DS} = V_{GS} - V_t$ for saturation; no channel below V_t.
2	18-19 2(a)	6	Threshold voltage	Inversion: surface potential moves by $-2\phi_F$; depletion depth and charge $Q_{B0} = -\sqrt{2qN_A\varepsilon_{Si}\lvert 2\phi_F\rvert}$; add work-function difference, $-2\phi_F$, $-Q_{B0}/C_{ox}$ and $-Q_{ox}/C_{ox}$.
3	18-19 TT2 8(a)	5	Threshold voltage	With V_{SB} the surface must move by $-2\phi_F + V_{SB}$; depletion charge $-\sqrt{2qN_A\varepsilon_{Si}\lvert -2\phi_F + V_{SB}\rvert}$; threshold as the sum of four terms; subtract V_{T0}; $V_T = V_{T0} + \gamma(\sqrt{\lvert -2\phi_F + V_{SB}\rvert} - \sqrt{\lvert 2\phi_F\rvert})$.
CMOS inverter				
4	19-20 3(a)	7	DC characteristics	Equal nMOS and pMOS currents with the right region for each; B: quadratic, smaller root; D: quadratic, smaller root; C: both saturated gives V_M; slope -1 in B and D gives V_{IL}, V_{IH}.
5	18-19 2(d)	8	DC characteristics	Equal nMOS and pMOS currents with the right region for each; B: quadratic, smaller root; D: quadratic, smaller root; C: both saturated gives V_M; slope -1 in B and D gives V_{IL}, V_{IH}.
Fabrication				
6	18-19 1(b)	5	Oxidation	Three fluxes $h(C^* - C_O)$, $D(C_O - C_I)/x_O$, $k_S C_I$; equal in steady state; eliminate C_O, C_I; $dx_O/dt = F/N_1$; separate and integrate; $x_O^2 + Ax_O = B(t + \tau)$; linear for thin, parabolic for thick oxide.
7	18-19 TT1 7(b)	6	Oxidation	Three fluxes $h(C^* - C_O)$, $D(C_O - C_I)/x_O$, $k_S C_I$; equal in steady state; eliminate C_O, C_I; $dx_O/dt = F/N_1$; separate and integrate; $x_O^2 + Ax_O = B(t + \tau)$; linear for thin, parabolic for thick oxide.

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No.	Paper	Marks	Derivation	Outline
Delay and logical effort				
8	18-19 3(d)	7	Best number of stages	Equal stage efforts: $D = NF^{1/N} + P$; added inverters add p_{inv} each; differentiate $NF^{1/N}$; $p_{inv} + \rho(1 - \ln \rho) = 0$; $\rho = 3.59$; $\hat{N} = \log_\rho F \approx \log_4 F$.
Power				
9	20-21 3(c)	8	Dynamic power	Supply gives $Q = C_L V_{DD}$ at V_{DD}: $C_L V_{DD}^2$ per cycle; half stored, half lost, then the stored half lost on discharge; $P = \alpha C_L V_{DD}^2 f$; saturated nMOS current; $t_{pd} = C_L(V_{DD}/2)/I_{sat} \approx C_L/(\beta V_{DD})$.
10	19-20 5(d)	6	Leakage	Subthreshold law $I = I'_{off} 10^{(V_{GS}+\eta V_{DS})/S}$; bottom: $V_{GS} = 0$, $V_{DS} = V_x$; top: $V_{GS} = -V_x$, $V_{DS} = V_{DD} - V_x$; equal currents give V_x; divide by the single-transistor leakage.

7. Design recipes:

Every design asked, in book order. Each recipe gives the steps in the order to take them; the link opens the full working.

No.	Paper	Marks	Design	Recipe
Delay and logical effort				
1	20-21 2(c)	7	Path delay	$H = 4$; list configurations (NAND8 + INV, NAND4 + NOR2, NAND2 + NOR4, four-stage trees); for each G, P, $F = GH$, $D = NF^{1/N} + P$; pick the smallest.
2	19-20 3(c)	7	Path delay	G, $B = 8$, $H = 9.6$, $F = 153.6$; $\hat{N} = \log_4 F = 3.63$; polarity needs an odd number of added inverters, so 3 stages (add 1 inverter); $\hat{f} = F^{1/3}$; $\hat{D} = 3\hat{f} + P$; size backwards.
Combinational logic styles				
3	20-21 3(a)	0	Static gates	Truth table: $Y = KB\overline{D} + K\overline{B}D = K(B \oplus D)$; nMOS network $K(B\overline{D} + \overline{B}D)$ under $\overline{Y}$ plus an inverter; implement in static CMOS, nMOS with depletion load, pseudo-nMOS and dynamic (domino) styles.
4	18-19 TT2 8(c)	5	Static gates	Build $\overline{Y}$: nMOS (A–B series) in parallel with [C in series with (A parallel B)]; pMOS is the dual; add an inverter for Y (12 transistors).
Sequential circuits and timing				
5	20-21 6(c)	6	FSM	States S0 (y = 1), S1, S2 in a ring; encode 00, 01, 10 (11 don't care); state table; next-state $S'_1 = S_0$, $S'_0 = \overline{S_1 + S_0}$; $y = \overline{S_1}\,\overline{S_0}$; two D flip-flops and one NOR2.
Datapath				
6	18-19 6(b)	6	Adders	$S = A \oplus B \oplus C_{in}$, $C_{out} = AB + C_{in}(A \oplus B)$; build from two half adders and an OR, or the 28-transistor mirror adder with $S = ABC_{in} + (A + B + C_{in})\overline{C_{out}}$.
Verilog, VHDL and design flow				
7	20-21 6(b)	5	Verilog	Segment names a to g; 'always @(*) case (data)' mapping 0 to 9 to the 7-bit patterns (0: 1111110, ..., 9: 1111011); default blank.
8	20-21 6(d)	3	Verilog	State register 'always @(posedge clk, posedge reset)'; next-state 'case' (S0 → S1 → S2 → S0); 'assign y = (state == S0)'.
9	18-19 1(d)	8	VHDL	Simplify: $F = AB + \overline{A}BC + BC = B(A + C)$; entity with inputs A, B, C and output F; architecture with one concurrent assignment 'F <= B and (A or C);'.

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8. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

No.	Paper	Marks	Subtopic	Method	Answer
Delay and logical effort					
1	20-21 1(d)	5	Path delay	Logical effort $G = \prod g$, branching $B = \prod b$, electrical $H = C_{out}/C_{in}$; $F = GBH$; $\hat{f} = F^{1/N}$; $\hat{D} = N\hat{f} + P$; size backwards from the load: $C_{in} = gC_{out}/\hat{f}$.	$F = 125$, $\hat{f} = 5$, $P = 7$, $\hat{D} = 22\tau$; $x = 10\lambda$, $y = 15\lambda$
2	20-21 2(a)	5	Elmore delay	Elmore delay to a node: sum over every resistor on the path from the input of R times all the capacitance downstream of it.	$t_{D7} = R_1(C_1 + \cdots + C_8) + R_6(C_6 + C_7 + C_8) + R_7(C_7 + C_8)$; t_{D5} similarly along its path
3	20-21 3(b)	4	Sizing	Unit inverter n 1, p 3 ($\mu_n = 3\mu_p$); series transistors need double width, parallel ones unit width; find the worst path in each network and size each device.	nMOS: A = 3/2, B = C = D = 3; pMOS: D = 3, A = B = C = 6
4	19-20 2(b)	7	Gate delay	Unit NAND3 (nMOS 3, pMOS 2); output and internal node capacitances; Elmore delay for the worst (propagation) and best (contamination) cases, rising and falling.	$t_{pdr} = 29RC$, $t_{pdf} = 32RC$, $t_{cdr} = 9.67RC$, $t_{cdf} = 29RC$
5	19-20 3(b)	6	Path delay	Logical effort $G = \prod g$, branching $B = \prod b$, electrical $H = C_{out}/C_{in}$; $F = GBH$; $\hat{f} = F^{1/N}$; $\hat{D} = N\hat{f} + P$; size backwards from the load: $C_{in} = gC_{out}/\hat{f}$.	$F = 116.7$, $\hat{f} = 3.29$, $\hat{D} = 20.1\tau$; $x = 2.82$, $y = 1.85$, $z = 14.6$
6	18-19 3(b)	6	Gate delay	Size the NOR3 for unit resistance (pMOS series 6, nMOS 1); node capacitances: output $(9 + 7h)C$, internal $12C$; Elmore delay from the worst-case RC ladders, rising and falling.	$t_{pdr} = (21 + 7h)RC$, $t_{pdf} = (9 + 7h)RC$
7	18-19 3(c)	10	Path delay	Logical effort $G = \prod g$, branching $B = \prod b$, electrical $H = C_{out}/C_{in}$; $F = GBH$; $\hat{f} = F^{1/N}$; $\hat{D} = N\hat{f} + P$; size backwards from the load: $C_{in} = gC_{out}/\hat{f}$.	$G = 2.96$, $B = 34.63$, $H = 2.6$, $F = 266.8$, $f = 4.04$, $P = 8$, $D = 24.2\tau$; $x = 4.46$, $y = 8.84$, $z = 16.1$
8	18-19 TT2 8(d)	5	Path delay	Logical effort $G = \prod g$, branching $B = \prod b$, electrical $H = C_{out}/C_{in}$; $F = GBH$; $\hat{f} = F^{1/N}$; $\hat{D} = N\hat{f} + P$; size backwards from the load: $C_{in} = gC_{out}/\hat{f}$.	$G = 25/9$, $B = 91/16$, $H = 8$, $F = 126.4$, $f = 3.35$, $P = 7$, $D = 20.4\tau$; $x = 2.87$, $y = 1.78$, $z = 14.3$
9	18-19 TT2 8(e)	5	Best number of stages	$p_{inv} + \rho(1 - \ln\rho) = 0$ gives $\rho \approx 3.59$; $\hat{N} = \ln F/\ln\rho$; round to a whole number; $f = F^{1/N}$.	$\hat{N} = 3.79$, so $N = 4$ with $f = 3.35$
Power					
10	19-20 4(c)	4	Dynamic power	C = capacitance per mm^2 × area; $P = \alpha C V_{DD}^2 f$.	about 1.15 W
11	18-19 4(c)	7	Activity factor	$P(\text{output} = 1)$ for each gate from the input probabilities; activity $\alpha = P(1 - P)$ (random independent data).	Tree: n_1, n_2: 3/16, Y: 15/256. Chain: n_3, n_4: 3/16; n_5, n_6: 7/64; n_7, Z: 15/256
12	18-19 4(d)	8	Dynamic power	C = transistor width × capacitance per μm for logic and memory; $P_{sw} = \alpha C V_{DD}^2 f$ for each; static: subthreshold and gate leakage per μm × total width × V_{DD}.	switching 6.1 W; static 859 mW (14 %)

No.	Paper	Marks	Subtopic	Method	Answer
Combinational logic styles					
13	20-21 1(b)	8	Pseudo-nMOS	Body tied to source, so $V_{SB} = 0$; nMOS linear, pMOS saturated at $V_{in} = 5$ V, $V_{out} = 0.25$ V; equate the currents to get k_n/k_p and the width ratio.	$k_n/k_p = 8.26$; $(W/L)_n/(W/L)_p = 3.44$
14	19-20 5(c)	9	Static gates	$H = C_{out}/C_{in}$; NAND–NAND: $G = (4/3)^2$, $P = 4$; AOI22 + inverter: $G = 2$, $P = 5$; $\hat{D} = 2F^{1/2} + P$ for each; size the gates.	NAND–NAND 9.96τ (faster); AOI22 + inverter 11.3τ
Sequential circuits and timing					
15	20-21 4(a)	10	Setup and hold	Clock arrival window at each flip-flop; data arrival window at FF2; hold: earliest data $\geq$ latest clock + hold; setup: latest data + setup $\leq T$ + earliest clock.	no hold violation (slack +1 ns); no setup violation if $T \geq 21$ ns
16	19-20 5(b)	4	Cycle time	Add the delays along the self-bypass loop; $T_c = t_{pcq} + t_{pd} + t_{setup}$; hold check: $t_{ccq} + t_{cd} \geq t_{hold}$.	$T_c = 1152$ ps (about 0.87 GHz); no hold failure

9. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

No.	Paper	Kind	Diagram
MOS transistor theory			
1	20-21 1(a)	Diagram	accumulation, depletion and inversion on p-type Si
2	20-21 1(a)	Graph	C-V curve and the gate capacitance by region
3	19-20 1(d)	Graph	electron energy along long and short channels
4	19-20 1(d)	Graph	drain current on a log scale against gate voltage
5	19-20 2(a)	Diagram	nMOS in cutoff, linear and saturation
6	19-20 2(a)	Graph	I_{DS} against V_{DS} and V_{GS}
7	18-19 2(a)	Diagram	accumulation, depletion and inversion under the gate
8	18-19 2(a)	Graph	threshold voltage built up from its four parts
9	18-19 TT2 8(a)	Graph	threshold voltage against source-to-body voltage
10	18-19 TT2 8(b)	Graph	electron energy along long and short channels
11	18-19 TT2 8(b)	Diagram	electron wave tunnelling through a thin oxide
CMOS inverter			
12	20-21 2(b)	Graph	VTC with unity-gain points for several beta ratios
13	20-21 3(d)	Circuit	unskewed, HI-skew and LO-skew inverters with widths
14	19-20 3(a)	Graph	nMOS and pMOS currents against V_{out}
15	19-20 3(a)	Graph	VTC with the five regions
16	18-19 2(c)	Circuit	saturated-load nMOS inverter
17	18-19 2(c)	Graph	its transfer curve
18	18-19 2(d)	Graph	nMOS and pMOS currents against V_{out} for several inputs
19	18-19 2(d)	Graph	VTC with the five regions
Fabrication			
20	20-21 1(e)	Graph	dopant profiles of diffusion and implantation
21	20-21 5(c)	Diagram	positive and negative resist with a mask
22	20-21 5(d)	Diagram	process sequence for patterning SiO_2

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No.	Paper	Kind	Diagram
23	19-20 1(a)	Diagram	isotropic and anisotropic etch profiles
24	19-20 1(a)	Diagram	plasma etcher and the etch steps
25	19-20 1(a)	Graph	etch rates against O_2 and H_2 added
26	19-20 1(b)	Diagram	the RCA clean sequence
27	19-20 1(c)	Diagram	photolithography steps with a positive resist
28	18-19 1(a)	Diagram	parallel-plate plasma etcher and the etch steps
29	18-19 1(a)	Graph	Si and SiO_2 etch rates against O_2 and H_2 added
30	18-19 1(b)	Graph	oxidant concentration from gas through oxide to silicon
31	18-19 1(b)	Graph	oxide thickness against time with the linear and parabolic limits
32	18-19 1(c)	Diagram	the loop that sustains a plasma
33	18-19 TT1 7(a)	Diagram	photolithography steps with a positive resist
34	18-19 TT1 7(b)	Graph	oxidant concentration from gas through oxide to silicon
35	18-19 TT1 7(b)	Graph	oxide thickness against time with the linear and parabolic limits
36	18-19 TT1 7(c)	Diagram	CVD epitaxy reactor and the crystal continuing
Delay and logical effort			
37	20-21 1(d)	Circuit	the path with the chosen sizes
38	20-21 2(a)	Circuit	RC tree of Figure 2(a)
39	20-21 2(c)	Circuit	five configurations with their delays
40	20-21 3(b)	Circuit	the sized gate
41	19-20 2(b)	Circuit	unit NAND3 and its RC models
42	19-20 3(b)	Circuit	the path with the chosen sizes
43	19-20 3(c)	Circuit	decoder path with the added inverter
44	19-20 3(d)	Circuit	NAND2 with outer and inner inputs
45	18-19 3(b)	Circuit	NOR3 with widths and its Elmore equivalents
46	18-19 3(c)	Circuit	the path with the input capacitance of every gate
47	18-19 3(d)	Graph	path delay against the number of stages
48	18-19 TT2 8(d)	Circuit	the path with the input capacitance of every gate
49	18-19 TT2 8(e)	Graph	path delay against the number of stages
Power			
50	20-21 2(d)	Diagram	charging and discharging the load capacitance
51	20-21 3(c)	Graph	dynamic power and delay against V_{DD}
52	19-20 4(b)	Waveform	a glitch in an XOR gate
53	19-20 5(d)	Circuit	one OFF transistor and two in series
54	18-19 4(b)	Diagram	energy from the supply stored and dissipated on rising and falling outputs
55	18-19 4(c)	Diagram	the tree and chain with each node's probability and activity
Interconnect			
56	19-20 4(d)	Diagram	two neighbouring wires with coupling and ground capacitances
57	19-20 4(e)	Diagram	sheet resistance of a square of a layer
Scaling			
58	19-20 2(c)	Graph	transistors per chip against year
59	19-20 4(a)	Diagram	a transistor before and after scaling
60	18-19 2(b)	Graph	transistors per chip against year
61	18-19 2(b)	Diagram	a transistor before and after scaling

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No.	Paper	Kind	Diagram
Combinational logic styles			
62	20-21 1(b)	Graph	pseudo-nMOS transfer curves
63	20-21 1(b)	Circuit	pseudo-nMOS inverter
64	20-21 1(c)	Circuit	schematic of $\overline{(A+B+C)D}$
65	20-21 1(c)	Diagram	stick diagram
66	20-21 2(e)	Circuit	domino OR gate
67	20-21 2(e)	Waveform	precharge and evaluate
68	20-21 3(a)	Circuit	static CMOS implementation
69	20-21 3(a)	Circuit	depletion-load and pseudo-nMOS versions
70	20-21 3(a)	Circuit	footed and unfooted dynamic versions
71	20-21 4(d)	Circuit	two-input AND with transmission gates and with pass transistors
72	20-21 6(a)	Circuit	2-to-1 multiplexer with pass transistors and with transmission gates
73	19-20 2(d)	Circuit	general CVSL gate
74	19-20 5(c)	Circuit	unit AOI22 with widths
75	19-20 5(c)	Circuit	the two designs with input capacitances
76	19-20 6(a)	Circuit	dynamic NAND2 and its waveforms
77	19-20 6(b)	Circuit	OAI22 and inverter symbolic diagram
78	19-20 6(b)	Diagram	stick diagram
79	18-19 4(a)	Graph	output against time for nMOS and pMOS pull-ups
80	18-19 5(a)	Circuit	NAND3 symbolic diagram with truth table
81	18-19 5(a)	Diagram	NAND3 stick diagram
82	18-19 6(a)	Circuit	footed dynamic inverter
83	18-19 6(a)	Waveform	precharge and evaluate over two cycles
84	18-19 TT2 8(c)	Circuit	CMOS compound gate with widths and the output inverter
Sequential circuits and timing			
85	20-21 4(a)	Waveform	setup and hold window
86	20-21 4(a)	Diagram	timeline of the arrivals
87	20-21 4(b)	Circuit	asynchronous resettable flip-flop
88	20-21 4(b)	Waveform	timing with reset
89	20-21 6(c)	Diagram	state transition diagram
90	20-21 6(c)	Circuit	FSM with two D flip-flops and a NOR gate
91	20-21 6(f)	Waveform	the four rows; Q changes only when ϕ rises
92	19-20 5(a)	Waveform	flip-flop to flip-flop min-delay timing
93	19-20 5(b)	Diagram	the ALU self-bypass path
94	18-19 5(c)	Waveform	max-delay and min-delay timing between two flip-flops
95	18-19 5(d)	Circuit	flip-flop with synchronous reset
96	18-19 5(d)	Waveform	timing with reset
Datapath			
97	20-21 5(b)	Block diagram	two-bus and three-bus datapaths
98	18-19 6(b)	Circuit	full adder from XOR, AND and OR gates
99	18-19 6(b)	Circuit	mirror adder
Memories and arrays			
100	20-21 5(a)	Circuit	6T SRAM cell
101	20-21 5(a)	Circuit	2×2 SRAM array with control signals
102	20-21 5(a)	Waveform	read timing
103	20-21 5(a)	Circuit	cross-coupled sense amplifier
104	20-21 6(e)	Diagram	PLA dot diagram of XOR
105	19-20 6(c)	Diagram	SRAM cell logic diagram
106	19-20 6(c)	Circuit	6T SRAM cell

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No.	Paper	Kind	Diagram
107	19-20 6(d)	Circuit	pseudo-nMOS ROM
108	19-20 6(d)	Diagram	ROM dot diagram
109	18-19 6(c)	Diagram	PLA dot diagram for s and c
110	18-19 6(d)	Circuit	1T DRAM cell and folded sub-array
111	18-19 6(d)	Waveform	read of a stored 1
Verilog, VHDL and design flow			
112	20-21 4(c)	Diagram	design flow from Verilog to a packaged chip
113	20-21 6(b)	Diagram	segment names and the pattern of each digit
114	18-19 1(d)	Diagram	the expression and its simplified gate form

10. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

Pair	Key difference	Asked in
CMOS inverter		
Saturated-load nMOS vs CMOS inverter	ratioed, $V_{OH} = V_{DD} - V_T$, static power vs ratioless, full swing, no static power	18-19 2(c)
Fabrication		
Wet vs dry etching	isotropic, selective, cheap vs anisotropic, precise, less selective	19-20 1(a)
Diffusion vs ion implantation	thermal, peak at the surface, less control vs accelerated ions, buried peak, precise, needs anneal	20-21 1(e)
Positive vs negative photoresist	exposed part dissolves, same image vs exposed part stays, reverse image	20-21 5(c)
Power		
Dynamic vs static power	$\alpha C V_{DD}^2 f$ when switching vs leakage $I_{static} V_{DD}$ all the time	20-21 2(d), 18-19 4(b)
Scaling		
Constant-field vs constant-voltage scaling	voltages scale too, power density constant vs voltage fixed, faster but power density rises	19-20 4(a), 18-19 2(b)
Combinational logic styles		
nMOS vs pMOS pull-up	passes a weak 1 ($V_{DD} - V_t$) vs passes a strong 1	18-19 4(a)
Static vs dynamic CMOS	full complementary networks, no clock vs one precharge pMOS, clocked, fewer transistors, faster, needs monotonic inputs	19-20 6(a), 18-19 6(a)
Transmission gate vs nMOS pass transistor	full swing, two transistors vs one transistor, weak 1	20-21 4(d)
Sequential circuits and timing		
Max-delay vs min-delay constraint	data must arrive before setup, fixed by a slower clock vs must not arrive before hold, fixed only by redesign	18-19 5(c)
Synchronous vs asynchronous reset	acts at the next clock edge vs acts at once, independent of the clock	20-21 4(b), 18-19 5(d)
Memories and arrays		
PLA vs PAL	both planes programmable vs AND programmable, OR fixed	18-19 6(c)
DRAM vs SRAM cell	1T1C, dense, refresh, destructive read vs 6T latch, fast, no refresh	18-19 6(d)
SRAM read vs write	pull-down stronger than access, no flip vs access stronger than pull-up, cell flips	19-20 6(c)
Verilog, VHDL and design flow		
VHDL vs Verilog	strongly typed, verbose vs C-like, compact	18-19 3(a)

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11. Repeated questions:

7 questions came back, exactly or with new numbers. Most repeated first; the latest paper first in each row.

No.	Times	Topic	Subtopic	Asked in (marks)
1	2	Scaling	Scaling rules	19-20 4(a) (5), 18-19 2(b) (6)
2	2	CMOS inverter	DC characteristics	19-20 3(a) (7), 18-19 2(d) (8)
3	2	Combinational logic styles	Stick diagrams	19-20 6(b) (6), 18-19 5(a) (8)
4	2	Combinational logic styles	Dynamic logic	19-20 6(a) (6), 18-19 6(a) (5)
5	2	Fabrication	Lithography	19-20 1(c) (8), 18-19 TT1 7(a) (8)
6	2	Delay and logical effort	Path delay	19-20 3(b) (6), 18-19 TT2 8(d) (5)
7	2	Fabrication	Oxidation	18-19 1(b) (5), 18-19 TT1 7(b) (6)

12. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical	Diagram	Design
MOS transistor theory	3	3	·	·	·
CMOS inverter	4	2	·	·	·
Fabrication	10	2	·	·	·
Delay and logical effort	1	1	9	·	2
Power	3	2	3	·	·
Interconnect	2	·	·	·	·
Scaling	3	·	·	·	·
Combinational logic styles	5	·	1	6	2
Sequential circuits and timing	3	·	2	2	1
Datapath	1	·	·	·	1
Memories and arrays	2	·	·	4	·
Verilog, VHDL and design flow	2	·	·	·	3

13. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Delay and logical effort	79	16 %
2	Combinational logic styles	72	31 %
3	Fabrication	67	45 %
4	Power	48	54 %
5	Sequential circuits and timing	48	64 %
6	Memories and arrays	40	72 %
7	CMOS inverter	38	80 %
8	MOS transistor theory	33	87 %
9	Verilog, VHDL and design flow	24	92 %
10	Scaling	17	95 %
11	Datapath	13	98 %
12	Interconnect	10	100 %

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